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The Toda lattice, Dynkin diagrams, singularities and Abelian varieties*

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The periodic $l+1$ particle Toda lattices [6, 22] corresponding to extended Dynkin diagrams [7, 15–17] have $l+1$ polynomial constants of the motion, namely as many as there are dots in the Dynkin diagram. For most values of the constants, their intersection defines an l-dimensional affine invariant manifold which completes into a complex algebraic torus (Abelian variety [12]) by glueing on a divisor (l-1-dimensional submanifold) entirely specified by the extended Dynkin diagram: each point of the diagram goes canonically to a component of the divisor and each subdiagram governs the intersection of the corresponding divisors (Theorem 2). Therefore the global geometry of the complex invariant tori (Abelian varieties), such as polarization, dimension of certain linear systems, divisor equivalences, etc...., is entirely given by the extended Dynkin diagram (Theorem 1). In particular, an explicit geometric description for the case of three-particle Toda lattices is given in Theorem 4, which describes the divisors, their singularities and how the various divisors intersect.

It follows that the singularities of the divisor are canonically associated to semi-simple Dynkin diagrams. We spell this out: the divisor $D=\sum_0^l D_i$ consists of $l+1$ irreducible components D_i, each associated with a root α_i of the Dynkin diagram Δ. The intersection of k components $D_{i_1},\dots,D_{i_k}$ satisfies the following relation:

$$\left(\begin{array}{c}\text{the intersection multiplicity of}\\ \text{the intersection of } k \text{ components}\\ \text{of the divisor}\end{array}\right)=\frac{\text{order}(W)}{\text{determinant }(A)},\tag{0.1}$$

where W and A are the Weyl group and the Cartan matrix going with the subDynkin diagram $\alpha_{i_1},\dots,\alpha_{i_k}$ associated with the k components (Theorem 2).

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The intersection of all the divisors is empty, whereas the intersection of all divisors but one is a discrete number of points; we found the following expression for the set-theoretical number of points in terms of the Dynkin diagram

$$\#\left(\bigcap_{\beta\neq\alpha} D_\beta\right)=\frac{p_\alpha}{p_0}\,\frac{\text{order (Weyl group of the Dynkin diagram } \Delta\backslash\alpha_0)}{\text{order (Weyl group of the Dynkin diagram } \Delta\backslash\alpha)},$$

where the integers p_α are given by the null vector of the Cartan matrix going with Δ. In Theorems 1 and 2 we give the genus of the divisors D_α and the linear equivalences between them, all expressed in Lie theoretical terms. It is striking that the singularities of each component occur always at the intersections with other components, and nowhere else. That is to say the following inclusion holds for the singular locus $\operatorname{sing}(D_k)$ of D_k:

$$\operatorname{sing}(D_k)\subseteqq D_k\cap \sum_{\substack{0\leqq i\leqq l\\ i\neq k}} D_i,\qquad k=0,\dots,l$$

Also the multiplicity of the singularity of a particular component D_k, at its intersection with m other divisors, i.e.

$$\operatorname{sing}(D_k)\cap(D_{i_1}\cap\dots\cap D_{i_m}),\qquad \text{all } i_1,\dots,i_m\neq k$$

is entirely specified by the way the corresponding root α_k sits in the subDynkin diagram $\alpha_k,\alpha_{i_1},\dots,\alpha_{i_m}$. For instance, two arbitrary divisors D_{α_1} and D_{α_2} intersect generically according to the four patterns of table 1 (Theorem 2), depending on whether the roots α_1 and α_2 have 0, 1, 2 or 3 links. The divisors D_{α_1} will be smooth along most of the intersection $D_{\alpha_1}\cap D_{\alpha_2}$, when the roots α_1 and α_2 are connected with 0 or 1 link; D_{α_1} will experience a normal crossing along most of $D_{\alpha_1}\cap D_{\alpha_2}$, when α_1 and α_2 are connected with 2 links and the arrow is pointing away from α_1; D_{α_1} will osculate itself to order 4, when α_1 and α_2 are connected with 3 links and the arrow is pointing away from α_1. Consider the case where the arrow is pointing towards α_1: then the divisor D_{α_1} experiences a normal crossing along most of $D_{\alpha_1}\cap D_{\alpha_2}$, when 3 links connect α_1 and α_2, and D_{α_1} is smooth when 2 links connect α_1 and α_2.

Consider now the standard Toda lattice, i.e., the one associated with the Dynkin diagram of the Lie algebra $\mathscr{a}_l^{(1)}$. The tori are Jacobians of (spectral) hyperelliptic curves $\mathscr{C}$, as is well known. The divisors D_i are all translates of the theta-divisor Θ, implying that

$$\operatorname{sing}\Theta\subset\Theta\cap[(\Theta+(p-q))+(\Theta-(p-q))]$$

for any $p,q\in C$ such that p and q are interchanged by the hyperelliptic involution, and $p-q\in \operatorname{Jac}(C)$ is an $l+1$-torsion point, i.e., in the Jacobian $(l+1)(p-q)\equiv 0$. In this context, the multiplicity of the singular locus of the hyperelliptic Θ-divisor, – of ccourse provided by the Riemann-Kempf theorem – acquires a Lie theoretical meaning.

The techniques used to analyze the singularities hinge on the (explicit) Laurent solutions (Painlevé solutions) of the Toda equations, as discussed in[1], which also turn out to be precisely indexed by subDynkin diagrams (Theorem 3). Indeed there are as many families of Laurent solutions as subsets $\phi\neq\Delta'\subsetneqq\Delta$ and for each Δ' its leading behavior is specified by

$$2\varrho_{\hat{\Delta}'}=\begin{pmatrix}\text{the sum of the positive roots of the Lie algebra}\\ \text{associated with the dual subDynkin diagram}\end{pmatrix};$$

for precise statements, see (1.28). This has been observed by Flaschka and Zeng [11] for the finite non-periodic Toda lattice.

These facts lead to yet another way to associate singularities to Lie algebras. Classically, point singularities are produced by acting with a discrete group on an affine space and by taking the quotient; then the Dynkin diagram describes the intersection pattern of the exceptional divisors obtained by the process of blowing up the variety at the singularity. It leads us to suspect a connection between the two points of view: this classical correspondence and the one encountered in the context of the Periodic Toda lattice. This is unknown although an approach may be based on the following observation.

It is well known (see [3]) that this Toda equations can be given in terms of a splitting of the Kac-Moody Lie algebra, going with this extended Dynkin diagram and that the solutions can be written in terms of a Birkhoff factorization problem of this corresponding loop group [26, 27, 29]. As the generic factorization is associated with the main cell, the Laurent solutions correspond to a factorization not occuring in the main cell [10]; they are labeled by elements of the Weyl group of the Kac-Moody Lie algebra. Therefore, it is natural to conjecture that the singularities studied here are related to those appearing in the Birkhoff cells. Since the Toda lattice only deals with tridiagonal matrices, it is natural to expect that a similar study with wider band matrices should reveal the complete picture of the Birkhoff cell structure.

The results were presented at MSRI in June 1989, and subsequently H. Flaschka and L. Haine kindly provided a proof of our formula (0.1) for the non-periodic Toda lattice – a limiting case of the periodic Toda lattice –, using toric varieties techniques for G/B.

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§ 1. Introduction and results

Consider the $(l+1)\times(l+1)$ Cartan matrix [7, 17]

$$(1.1)\qquad A=\left(\frac{2\langle\alpha_i,\alpha_j\rangle}{\langle\alpha_j,\alpha_j\rangle}\right)_{0\leqq i,j\leqq l},$$

where $\alpha_1,\ldots,\alpha_l$ denote the simple roots of any of the simple Lie algebras $\mathscr{g}_0=\mathscr{a}_l,\mathscr{b}_l,\mathscr{c}_l,\mathscr{d}_l,\mathscr{e}_6,\mathscr{e}_7,\mathscr{e}_8,\mathscr{f}_4,\mathscr{g}_2$ and where α_0 is one of the three expressions:

$$(1.2)\qquad \alpha_0=\begin{cases}-(\text{highest long root of }\mathscr{g}_0)=-\sum_1^l q_i\alpha_i & (v=\mathrm{id})\\ -(\text{highest short root of }\mathscr{g}_0)=-\sum_1^l q_i'\alpha_i & (v\neq\mathrm{id})\\ -2(\text{highest short root of }\mathscr{b}_l)=-2\sum_l^l q_i'\alpha_i & (a_{2l}^{(2)})\end{cases}$$

where v is defined below. Clearly these extended Cartan matrices (1.1) are non invertible; each of them defines a Kac-Moody Lie algebra [15, 17]

$$L(\mathscr{g}, v)=\bigoplus s^j\mathscr{g}_{j\,\mathrm{m}^\circ\mathrm{d}\,m},$$

where s is a dummy variable and $\mathscr{g}$ is a simple Lie algebra with automorphism v of order m ($m=1, 2$ or 3); v is generated by an automorphism of the Dynkin diagram of $\mathscr{g}$ and $\mathscr{g}_j$ is the eigenspace associated with the eigenvalue $\exp(2\pi\sqrt{-1}\,j/m)$ of v. The eigenspace $\mathscr{g}_0$ going with eigenvalue 1 is a simple Lie algebra and agrees with the $\mathscr{g}_0$ discussed in (1.2).

This paper deals with the weight-homogeneous Hamiltonian system

$$(1.3)\qquad \dot{x}_i=x_iy_i,\quad \dot{y}=Ax,\quad \text{with } x,y\in\mathbb{R}^{l+1},\quad \langle p,y\rangle=0\,;$$

they are the equations for the Toda lattice in appropriate coordinates, with p the left null-vector of A ($pA=0$). This system whose variables x_i and y_i have weights[1] 2 and 1, can be written as a Lax pair

$$(1.4)\qquad \dot{L}=[L,P],$$

generated by the Kostant-Kirillov symplectic structure, where

$$L\equiv\sum_1^l y_i\hat{\lambda}_i-\sum_0^l x_ie_{\alpha_i}+\sum_0^l e_{-\alpha_i},\qquad P\equiv-\sum_0^l x_ie_{\alpha_i},$$

in terms of notation given below. The energy of the system is given by

$$(1.5)\qquad H=\frac{1}{2}\langle L,L\rangle=\frac{1}{2}\langle y,y\rangle-\sum_0^l\langle e_{\alpha_i},e_{-\alpha_i}\rangle x_i$$
$$=\frac{1}{2}\sum_{1\le i,j\le l}\langle\hat{\lambda}_i,\hat{\lambda}_j\rangle y_iy_j-\sum_0^l\frac{2x_i}{\langle\alpha_i,\alpha_i\rangle}.$$

Before stating the main theorems, we introduce some Lie-theoretical notation. If $\Delta'=\{\alpha_1,\dots,\alpha_n\}$ are the simple roots of any semi-simple Lie algebra, then the dual roots $(\hat{\alpha}_1,\dots,\hat{\alpha}_n)$, with

$$\hat{\alpha}_i\equiv\frac{2\alpha_i}{\langle\alpha_i,\alpha_i\rangle},$$

form a basis $\hat{\Delta}'$ of simple roots as well. The fundamental dominant weights $\lambda_1,\dots,\lambda_n$ (relative to the basis $\alpha_1,\dots,\alpha_n$) and dual weights $\hat{\lambda}_1,\dots,\hat{\lambda}_n$ are defined by

$$(1.6)\qquad \langle\lambda_i,\hat{\alpha}_j\rangle=\delta_{ij},\qquad\langle\hat{\lambda}_i,\alpha_j\rangle=\delta_{ij}.$$

[1] a system $\dot{z}=f(z), z\in\mathbb{R}^n$ is weight-homogeneous with weight $(z_i)=v_i+1$, when $f_i(\alpha^{v_1}z_1,\dots,\alpha^{v_n}z_n)=\alpha^{v_i+1}f_i(z)$. The invariants of such a system can always be taken weight-homogeneous

Let $\Lambda_r(\Delta')$ and $\Lambda_w(\Delta')$ be the integral lattices of the roots $\alpha_1, \ldots, \alpha_n$ and weights $\lambda_1, \ldots, \lambda_n$ respectively. If

$$\alpha_i = \sum_k m_{ik} \lambda_k ,$$

then

$$A_{ij} = \langle \alpha_i, \hat{\alpha}_j \rangle = \sum_k m_{ik} \langle \lambda_k, \hat{\alpha}_j \rangle = m_{ij}, \tag{1.7}$$

and thus

$$\alpha_i = \sum_k A_{ik} \lambda_k , \qquad \text{or equivalently } A_{\Delta'} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}. \tag{1.8}$$

The roots and weights all play an important role in the decomposition of the Lie algebras; in fact we have

$$\hat{\lambda}_i \in \mathscr{h} = \text{Cartan subalgebra} \subset \mathscr{g}_0 \subset L(\mathscr{g}, \nu)$$

$$\alpha_i \in \mathscr{h}^*$$

$$[h, e_\alpha] = \langle \alpha, h \rangle e_\alpha , \qquad h \in \mathscr{h}, \qquad e_\alpha = \text{root vector} \in L(\mathscr{g}, \nu)$$

$$[e_\alpha, e_{-\alpha}] = \hat{\alpha} \in \mathscr{h}, \qquad \langle e_\alpha, e_{-\alpha} \rangle = \frac{2}{\langle \alpha, \alpha \rangle}.$$

If we denote[2] by p and $\hat{p}$ the left- and right-null vectors of A, i.e.

$$pA = 0 \quad \text{and} \quad A\hat{p} = 0, \qquad p_0 = \hat{p}_0 \equiv 1, \tag{1.9}$$

then

$$\sum_0^l p_i \alpha_i = 0 \quad \text{and} \quad \sum_0^l \hat{p}_i \hat{\alpha}_i = 0$$

and

$$\delta_i \equiv \frac{p_i}{\hat{p}_i} = \frac{\langle \alpha_0, \alpha_0 \rangle}{\langle \alpha_i, \alpha_i \rangle}, \qquad i = 0, \ldots, l.$$

In this paper, Δ will always refer to the set $\{\alpha_0, \ldots, \alpha_l\}$, whereas $\Delta' \subsetneqq \Delta$ will be any subset of roots; introduce the notation

$$\Delta_\alpha = \Delta \backslash \alpha \quad \text{and} \quad \Delta_0 = \Delta \backslash \alpha_0 .$$

Observe the identity

$$\frac{p_\alpha \hat{p}_\alpha}{\det A_{\Delta_\alpha}} = \frac{1}{\det A_0}, \qquad \alpha \in \Delta . \tag{1.10}$$

Indeed, by Cramer's rule

$$p_\alpha = \frac{M_{\alpha 0}}{M_{00}} = \frac{M_{\alpha\alpha}}{M_{0\alpha}}, \qquad \hat{p}_\alpha = \frac{M_{0\alpha}}{M_{00}} = \frac{M_{\alpha\alpha}}{M_{\alpha 0}},$$

[2] all p_i and $\hat{p}_i$ are positive integers, except for $a_{2l}^{(2)}$, where $\hat{p} = (1, 1, \ldots, 1, 1/2)$

where $M_{\alpha\beta}$ is the determinant of the (α,β)th minor of A (with alternating sign convention understood) and so

$$\frac{\hat{p}_\alpha p_\alpha}{\det A_{\Delta_\alpha}}=\frac{\hat{p}_\alpha p_\alpha}{M_{\alpha\alpha}}=\frac{M_{\alpha 0}M_{0\alpha}}{M_{00}^2 M_{\alpha\alpha}}=\frac{1}{M_{00}}=\frac{1}{\det A_{\Delta_0}}.$$

Also define

$$2\varrho_{\Delta'}\equiv\sum(\text{positive roots generated by } \Delta') \tag{1.11}$$
$$=\sum_{\alpha\in\Delta'}\lambda_\alpha$$
$$=\sum_{\alpha\in\Delta'} n_\alpha \alpha .$$

Then we have

$$\langle 2\varrho_{\Delta'},\hat{\alpha}\rangle=1 \quad \text{for all} \quad \hat{\alpha}\in\hat{\Delta}'; \tag{1.12}$$

this relation defines $2\varrho_{\Delta'}$ uniquely.

Let $A_{\Delta'}$ be the Cartan matrix associated with the set Δ', let $\Sigma q_\alpha \alpha$ be the highest long root of Δ', let $|W_{\Delta'}|$ be the order of the Weyl group associated with Δ' and finally let m_i be the Weyl exponents[3]. Then we have the following relations[4]

$$|W_{\Delta'}|=\prod(m_i+1) \tag{1.13}$$
$$=l!\det(A_{\Delta'})\prod_{\alpha\in\Delta'} q_\alpha$$
$$=\frac{\prod_{\alpha\in\Delta'} n_\alpha}{\prod m_i}\det(A_{\Delta'}),$$

using the relation

$$l!\prod_{\alpha\in\Delta'} q_\alpha \prod m_i=\prod_{\alpha\in\Delta'} n_\alpha .$$

The equations (1.3) or their Lax pair (1.4) derive from a Kostant-Kirillov symplectic structure, as follows:

$$X_H: \dot{x}=x*\left(\delta^{-1}A_0{}^T\frac{\partial H}{\partial y}\right),\quad \dot{y}=-A_0\delta^{-1}\left(x*\frac{\partial H}{\partial x}\right) \tag{1.14}$$

with

$$x=(x_0,\dots,x_l)\in\mathbb{R}^{l+1},\qquad y=(y_1,\dots,y_l)\in\mathbb{R}^l,$$
$$x*x'=(x_0x_0',\dots,x_lx_l')\in\mathbb{R}^{l+1},\qquad \delta^{-1}=\text{diagonal}(\delta_0^{-1},\dots,\delta_l^{-1})$$
$$A_0=(A\text{ with the zeroth row removed}),\qquad \delta^0=\text{diagonal}(\delta_1,\dots,\delta_l);$$

the energy H, given by (1.5), reads, after multiplication by $\langle\alpha_0,\alpha_0\rangle/2$:

$$H=\frac{1}{2}\langle(A_{\Delta_0}^T)^{-1}\delta^0 y,y\rangle-\sum_0^l \delta_i x_i . \tag{1.15}$$

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[3] then the integers m_i+1 are the degrees of the basic invariant polynomials of the Weyl group $W_{\Delta'}$; there are $|\Delta'|$ of them

[4] The second and third relations hold only if Δ' is irreducible

The latter follows from the identity

$$(A_{\Delta_0}^T)^{-1}\delta^0 = \tfrac{1}{2}\langle\alpha_0,\alpha_0\rangle[\langle\hat{\lambda}_i,\hat{\lambda}_j\rangle]_{1\leq i,j\leq l},$$

itself an immediate consequence of (1.6) and

$$A_{\Delta_0}^T\begin{pmatrix}\hat{\lambda}_1\\ \vdots\\ \hat{\lambda}_l\end{pmatrix}=\begin{pmatrix}\hat{\alpha}_1\\ \vdots\\ \hat{\alpha}_l\end{pmatrix}.$$

Theorem 1 (The geometry of the invariant tori). *The system* (1.1), *defined by the extended Cartan matrix A associated with $L(\mathfrak{g}, v)$*, has besides the Casimir function

$$H_0=\prod_0^l x_i^{p_i},$$

l invariants $H_1,\ldots,H_l$ in involution of weighted degree $m_1+1,\ldots,m_l+1$ (which are extensions of the Weyl polynomials of Δ_0). They generate commuting Hamiltonian flows, for the symplectic structure (1.14), *and they are straight line motions on Abelian varieties*

$$T^l=\frac{\mathbb{C}^l}{\Lambda},$$

where Λ is the lattice generated by the $2l$ columns appearing in the period matrix Ω_{D_0}, itself induced by the polarization of D_0. In fact, Ω_{D_0} has the form

$$\Omega_{D_0}=(I\ \ Z),\qquad \textit{when the automorphism } v\neq \textit{identity}$$

and

(1.16)
$$\Omega_{D_0}=\left(\begin{matrix}\delta_1 & & \\ & \ddots & \\ & & \delta_l\end{matrix}\quad Z\right),\qquad \textit{when } v=\textit{identity}$$

where[5] *$\delta_i=p_i/\hat{p}_i$. In all cases*

$$\operatorname{Im} Z>0 \quad \textit{and} \quad Z=Z^T.$$

Moreover the affine invariant manifold $\mathscr{A}=\bigcap_0^l\{H_i=c_i, (x,y)\in\mathbb{C}^{2l+1}\}$ completes into an Abelian variety by adjoining a divisor $D=\sum_{\alpha\in\Delta} D_\alpha$

$$\mathscr{A}=\bigcap_0^l\{H_i=c_i\}=T^l\backslash\Big(\sum_{\alpha\in\Delta} D_\alpha\Big),$$

where the set

$$D_i=D_{\alpha_i}=\{x_i^{-1}=0 \textit{ on } T^l\}.$$

The divisor (x_i) of zeroes and poles of x_i is given by

(1.17)
$$\begin{pmatrix}(x_0)\\ \vdots\\ (x_l)\end{pmatrix}=A\begin{pmatrix}D_0\\ \vdots\\ D_l\end{pmatrix},$$

[5] remember $A\hat{p}=0$ and $A^Tp=0$

inducing the following linear equivalence[6]

(1.18) $$\gamma_i D_i \simeq \gamma_i \hat{p}_i D_0 ,$$

where γ_i is the smallest positive integer such that $\gamma_i \lambda_i \in \Lambda_r(\Delta_0)$. Consequently any divisor D_i would give the same period matrix Ω, up to a factor $\hat{p}_i$. Moreover

(1.19) $$\dim L\left(\sum_0^l r_i D_i\right) = \left(\sum r_i \hat{p}_i\right)^l \begin{cases} \prod_1^l \delta_i & \text{for order 1 automorphisms } v \\ 1 & \text{for order 2 and 3 automorphisms} \end{cases}$$

For all $\Delta' \subset \Delta$, the behavior of $D_\alpha, \alpha \in \Delta'$ near $\bigcap_{\alpha \in \Delta'} D_\alpha$, away from $D_\beta, \beta \notin \Delta'$, is governed by the Lie algebra with simple roots Δ'. In particular we have:

Theorem 2 (Singularities and Dynkin diagrams). *Along any component of $\bigcap_{\alpha \in \Delta'} D_\alpha$, we have the following formula for the intersection multiplicity*:

(1.20) $$\text{mult. } (D_{\alpha_1} \cdot D_{\alpha_2} \cdot D_{\alpha_3} \ldots D_{\alpha_k}, \alpha_i \in \Delta') = \frac{|W_{\Delta'}|}{\det(A_{\Delta'})},$$

where $W_{\Delta'}$ and $A_{\Delta'}$ denote the Weyl group and the Cartan matrix, associated with the roots $\Delta' \subset \Delta$. Moreover, we have the following formula for the number of points:

(1.21) $$\#\left(\bigcap_{\beta \neq \alpha} D_\beta\right) = \frac{p_\alpha}{p_0} \frac{|W_{\Delta_0}|}{|W_{\Delta_\alpha}|}$$

and the global intersection number

(1.22) $$\begin{aligned}(D_0 \cdot D_1 \cdot \ldots \cdot \hat{D}_\beta \cdot \ldots \cdot D_l) &= \frac{|W_{\Delta_0}|}{\det(A_{\Delta_0})} \frac{\hat{p}_0}{\hat{p}_\beta} \\ &= D_0^l \prod_{\substack{\alpha \in \Delta \\ \alpha \neq \beta}} \hat{p}_\alpha \\ &= (g(D_0) - l + 1) \prod_{\substack{\alpha \in \Delta \\ \alpha \neq \beta}} p_\alpha l!.\end{aligned}$$

Moreover[7]

(1.23) $$g(D_0) - l + 1 = \prod_{\beta \in \Delta_0} \frac{q_\beta}{\hat{p}_\beta}.$$

In particular, a transversal slice through a generic point of $D_{\alpha_1} \cap D_{\alpha_2}$ intersects D_{α_1} and D_{α_2} according to curves, which appear in table 1, where each of the pictures is controlled by the subDynkin diagram of $\{\alpha_1, \alpha_2\}$.

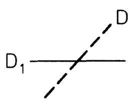
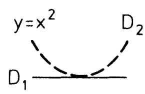
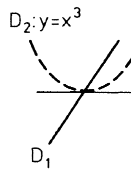
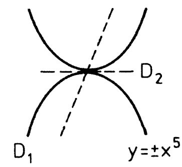
Remark. It is natural to conjecture the following generalization of (1.22) as an identity in *integral* homology:

$$D_{\alpha_1} \cdot D_{\alpha_2} \cdot \ldots \cdot D_{\alpha_k} \sim \left(\prod_i p_{\alpha_i}\right) \prod_{\beta \in \Delta_0} \left(\frac{q_\beta}{\hat{p}_\beta}\right) \cdot k! \, H_0^{(k)},$$

[6] Notice this is the best possible result. For the purpose of this statement, D_0 can be chosen to be any of the divisors D_α; then $\Lambda_r(\Delta_0)$ is replaced by $\Lambda_r(\Delta_\alpha)$

[7] $g(D_0)$ is the dimension of the space of holomorphic top forms on D_0

Table 1. Transversal slice through a generic point of $D_{\alpha_1} \cap D_{\alpha_2}$

diagram	$\underset{\alpha_1}{\cdot}\quad\underset{\alpha_2}{\cdot}$	$\underset{\alpha_1}{\cdot}$———$\underset{\alpha_2}{\cdot}$	$\underset{\alpha_1}{\cdot}\Longrightarrow\underset{\alpha_2}{\cdot}$	$\underset{\alpha_1}{\cdot}\Rrightarrow\underset{\alpha_2}{\cdot}$
Lie algebra	$a_1 \oplus a_1$	a_2	b_2	g_2
Singularity				
$\lvert W \rvert$	4	3!	8	12
$\det A$	4	3	2	1
inters. mult.	1	2	4	12
highest long root	$\alpha_1+\alpha_2$	$\alpha_1+\alpha_2$	$\alpha_1+2\alpha_2$	$2\alpha_1+3\alpha_2$
$2p=n_1\alpha_1+n_2\alpha_2$	$\alpha_1+\alpha_2$	$2\alpha_1+2\alpha_2$	$4\alpha_1+3\alpha_2$	$10\alpha_1+6\alpha_2$
$\lvert 2\varrho \rvert$	2	4	7	16
local equation	$yx=0$ $D_1 D_2$	$\underbrace{(y-x^2)}_{D_1}\underbrace{(y-ax^2)}_{D_2}=0$	$\overbrace{x(y-x^3)}^{D_1}\overbrace{(y-ax^3)}^{D_2}=0$	$\overbrace{(y-ax^5)(y-bx^5)}^{D_1}$ $\underbrace{(y-cx^5)x}_{D_2}=0$

with

$$H_0^{(k)} = \frac{D_0^k}{k!\prod_{\beta\in\Delta_0}(q_\beta/\hat{p}_\beta)}$$

being an *integral* cycle in homology.

In view of Theorem 1 in Adler and van Moerbeke [2], the Painlevé solutions of an algebraic integrable system form a coherent tree; the Painlevé solutions correspond to cells of different dimensions. D_α refers to the divisors of T^l, and we now recursively define $D_{\Delta'} \subset T^l$, for $\phi \neq \Delta' \subseteqq \Delta$, as follows:

$$D_{\Delta'} \equiv \left(\bigcap_{\alpha\in\Delta'} D_\alpha\right) \setminus \bigcap_{\Delta'' \supsetneqq \Delta'} D_{\Delta''} \tag{1.24}$$

with the understanding that

$$D_\Delta = \bigcap_{\alpha\in\Delta} D_\alpha \quad \text{and} \quad D_{\Delta\setminus\alpha} \equiv \bigcap_{\beta\neq\alpha} D_\beta \setminus D_\Delta . \tag{1.25}$$

For instance, $D_{\{\alpha\}}$ is the principal cell in D_α, i.e.

$$D_{\{\alpha\}} = D_\alpha \setminus \left(\bigcap_{\substack{\Delta'\ni\alpha \\ \Delta'\subseteq\Delta}} D_{\Delta'}\right) . \tag{1.26}$$

The next Theorem deals with the nature of the Laurent solutions $D_{\Delta'}(t)$ associated with each $\phi \neq \Delta' \subsetneqq \Delta$; the leading behavior is entirely specified by the element $\varrho_{\hat{\Delta}'}$. In the statement we need the following formulas

$$2\varrho_{\hat{\Delta}'} = \sum_{\hat{\alpha}' \in \hat{\Delta}} \hat{n}_\alpha \hat{\alpha}, \quad |2\varrho_\Delta| = \sum_{\alpha \in \Delta} n_\alpha, \quad -\langle 2\varrho_{\hat{\Delta}'}, \alpha \rangle = -\sum_{\beta \in \Delta'} A_{\alpha\beta} \hat{n}_\beta,$$

using the notations (1.11).

Theorem 3.I. (Painlevé solutions and Dynkin diagrams). *For the Hamiltonian system*

(1.27) $$X_\Delta : \dot{x}_i = x_i y_i \quad \text{and} \quad \dot{y} = A_\Delta x, \quad \langle p, y \rangle = 0,$$

there is a one-to-one correspondence between subsets $\Delta' \subsetneqq \Delta, 1 \leqq |\Delta'| \leqq l$, of simple roots and families $D_{\Delta'}(t)$ of Laurent solutions depending on $2l+1-|\Delta'|$ free parameters; they have the following form:

(1.28) $$x_\alpha(t) = t^{-2} \begin{Bmatrix} \hat{n}_\alpha + O(t^2) \quad \text{for} \quad \alpha \in \Delta' \\ b_\alpha t^{-\langle 2\varrho_{\hat{\Delta}}, \alpha \rangle + 2}(1 + c_\alpha t + O(t^2)) \quad \text{for} \quad \alpha \notin \Delta' \end{Bmatrix}$$

$$y_\alpha(t) = t^{-1} \begin{Bmatrix} -2 + O(t) \quad \text{for} \quad \alpha \in \Delta' \\ -\langle 2\varrho_{\hat{\Delta}'}, \alpha \rangle + c_\alpha t + O(t^2) \quad \text{for} \quad \alpha \notin \Delta' \end{Bmatrix}$$

with c_α such that

$$\sum_{\alpha \notin \Delta'} p_\alpha c_\alpha = 0 \quad \text{and} \quad -\langle 2\varrho_{\hat{\Delta}'}, \alpha \rangle = -\sum_{\beta \in \Delta'} A_{\alpha\beta} \hat{n}_\beta;$$

besides the $2(l - |\Delta'|) + 1$ free parameters b_α and c_α, there are $|\Delta'|$ other free parameters $d_1, \ldots, d_{|\Delta'|}$, occuring at the steps t^{m_i+1} in the brackets, where m_i are the Weyl exponents of the Lie algebra generated by Δ'. These expressions are weight-homogeneous, by assigning to t the weight -1 and to each free parameter a weight $k \geqq 0$, where the free parameter appears first within the brackets in the coefficient of t^k.

Moreover, for every $\phi \neq \Delta' \subsetneqq \Delta$, the following function determined by $\varrho_{\Delta'}$ has the following behavior on T^l:

(1.29) $$g_{\Delta'} \equiv \prod_{\alpha \in \Delta'} x_\alpha^{-n_\alpha/2} = \begin{cases} O(t) \quad \text{along} \quad D_{\{\alpha\}}(t), \alpha \in \Delta' \\ O(t^{|2\varrho|}) \quad \text{along} \quad D_{\Delta'}(t) \end{cases}$$

In most cases, the function (1.29) *is meromorphic on T^l and if not, its square will be.*

Notice that the assignment of weights to the variable t and the free parameters is consistent with the weights 2 and 1 of x_α and y_α. In particular, if $H(x, y)$ is a weight-homogeneous polynomial of degree M, then $H(x(t, b, c, d), y(t, b, c, d))$ has weight M, as well, viewed as an expression in the free parameters b, c, d and t.

In Theorem 3.II we show that the Laurent solutions $D_{\Delta'}(t)$ form a coherent tree, in the precise sense of Adler and van Moerbeke [2], i.e., they glue together according to the partial ordering $\Delta' \subset \Delta$. At the same time we indicate what these solutions mean in terms of the geometry of the divisors $D_\alpha \subset T^l$. In particular, in view of the coherence, we show how the lower balances $D_{\Delta'}(t)$ can be obtained from the principal ones $D_{\{\alpha\}}(t)$: for some variables, the lower balances are obtained by taking straight limits of the coefficients of t^k for an appropriate limit of the free parameters. For others, the coefficients of t^k would then blow up and thus something subtler must be done. According to the general theory (see [2]), the problem can be remedied by means of a birational transformation. Consequently the variables x and y can be meromorphically continued to T^l.

Theorem 3.II. (Coherence of Painlevé solutions). *The families $D_\Delta(t)$ obtained formally in Theorem 3.I relate to the affine varieties $D_{\Delta'}$ defined in* (1.25) *as follows (in terms of the flow φ^t of the vector field* (1.27)*)*

$$D_{\Delta'}(t)=\varphi^t(D_{\Delta'})$$

with

$$\text{dimension}(D_{\Delta'})=l-|\Delta'|. \tag{1.30}$$

They form a coherent tree, depicted in Fig. 1, with (for definitions and notations, see [2]*)*

$$\coprod_{\Delta''\supseteq\Delta'} D_{\Delta''}=\bigcap_{\alpha\in\Delta'} D_\alpha\,;$$

in particular

$$D_\Delta=\bigcap_{\alpha\in\Delta} D_\alpha=\phi\,.$$

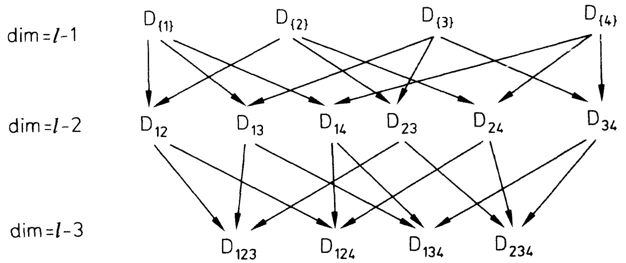


Fig. 1

The lower balances can be obtained from the principal ones by the recipe explained below. Consider the principal balances $D_{\{\alpha\}}(t), \alpha\in\Delta'$ and the lower one $D_{\Delta'}(t)$, $(1<|\Delta'|\leq l)$; we distinguish the following two cases:

(1) *$\beta\notin\Delta'$; for $\alpha\in\Delta'$ $x_\beta(t,D_{\{\alpha\}}(t))=$ a Taylor series in t and a holomorphic function in a T^l-neighborhood of $D_{\Delta'}$.*

$$x_\beta(t,D_{\{\alpha\}}(t))\to x_\beta(t,D_{\Delta'}(t)) \tag{1.31}$$

by taking an appropriate limit of the free parameters, term by term,

(2) *$\beta\in\Delta'$; for $\alpha\in\Delta'$* we have

$$\tilde{x}_\beta(t,D_{\{\alpha\}}(t))\equiv\frac{\dfrac{x_\beta}{g^2_{\Delta'}}(t,D_{\{\alpha\}}(t))}{\dfrac{1}{g^2_{\Delta'}}(t,D_{\{\alpha\}}(t))} \tag{1.32}$$

$$=\frac{\textit{holomorphic series}}{\textit{holomorphic series}}\ \textit{in } t \textit{ and in a } T^l\textit{-neighborhood of } D_{\Delta'}.$$

By taking an appropriate limit (depending on α) of the free parameters – term by term – in the numerator and the denominator, and then by dividing the two Taylor series in t, one obtains $x_\beta(t, D_{\Delta'}(t))$, i.e.

$$\tilde{x}_\beta(t, D_{\{\alpha\}}(t)) \to x_\beta(t, D_{\Delta'}(t)). \tag{1.33}$$

In both cases the series for $y_\beta(t, D_{\Delta'}(t))$ is obtained by using $y_\beta = \dot{x}_\beta / x_\beta$. Moreover the Laurent solutions satisfy the global completeness condition (see [2]):

$$\sum_{\phi = \Delta' \subsetneq \Delta} d_{\Delta'} = \frac{1}{2} |W_{\Delta_0}| \sum_0^l p_i,$$

where

$$d_{\Delta'} = \text{degree} \left\{ \begin{array}{l} l\text{-}1\text{-dimensional components of the image} \\ \text{of } (\tilde{x}(o,u), \tilde{y}(o,u)) \text{ in weighted projective space} \\ \mathbb{P}^{2l+1}_{1,2} \equiv \{(x,y,z) \sim (s^2 x, sy, sz)\} \end{array} \right\}$$

Remark. Intuitively, the global completeness guarantees that all affine subvarieties $D_{\Delta'}$ are obtained by blowing up or blowing down the natural closure $\bar{\mathscr{A}}$ of the invariant manifold $\mathscr{A}$ embedded into $\mathbb{P}^{2l+1}_{1,2}$, namely

$$\bar{\mathscr{A}} = \{H_0 = z^{2\Sigma p_i}\} \cap \{H_i = c_i z^{(m_i+1)}\} \subset \mathbb{P}^{2l+1}_{1,2},$$

along the hyperplane section $\bar{\mathscr{A}} \cap \{z = 0\}$.

Corollary 1 (principal and lowest balances). (i) *Solutions through D_α (principal balances): When $\Delta' = \{\alpha\}$, only x_α blows up among the x's and*

$$2\varrho_{\hat{\Delta}'} = \hat{\alpha}, \quad -\langle 2\varrho_{\hat{\Delta}'}, \beta \rangle = -A_{\beta\alpha}, \quad \hat{n}_\alpha = 1.$$

The Laurent solutions have $2l$ free parameters, namely the $2l$-1 free parameters b_β and c_β above and one extra parameter b_α appearing as follows:

$$D_{\{\alpha\}}: x_\beta = t^{-2} \begin{cases} 1 + b_\alpha t^2 + \ldots + x_\alpha^{(k)} t^k + \ldots & \text{for } \beta = \alpha \\ b_\beta t^{-A_{\beta\alpha}+2} + \ldots + x_\beta^{(k)} t^k + \ldots & \text{for } \beta \neq \alpha \end{cases} \tag{1.34}$$

$$y_\beta = t^{-1} \begin{cases} -2 + 2b_\alpha t^2 + \ldots + y_\alpha^{(k)} t^k + \ldots & \text{for } \beta = \alpha \\ -A_{\beta\alpha} + c_\beta t + \ldots + y_\beta^{(k)} t^k + \ldots & \text{for } \beta \neq \alpha, \end{cases} \qquad \sum_{\alpha \neq \beta} p_\alpha c_\alpha = 0$$

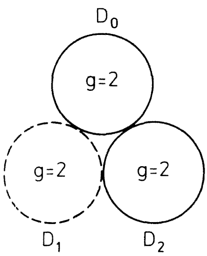
(ii) *Solutions through $\bigcap_{\alpha \neq \beta} D_\alpha$ (lowest balances): When $\Delta' = \Delta_\beta \equiv \Delta \backslash \{\beta\}$, all x_α, but x_β, blow up as t^{-2} and*

$$-\langle 2\varrho_{\hat{\Delta}_\beta}, \beta \rangle + 2 = 2 \sum_{\alpha \in \Delta} p_\alpha / p_\beta, \quad c_\beta = 0.$$

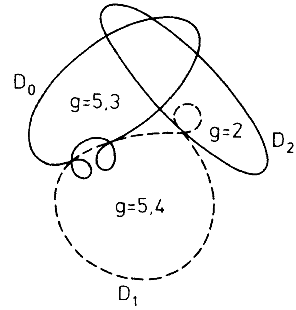
The series depend on the minimum number $l+1$ of parameters, accounting for the $l+1$ constants of motions: one free parameters is b_β and l free parameters enter at t^{m_i+1} where the m_i are the l Weyl exponents of the Lie algebra generated by Δ_β.

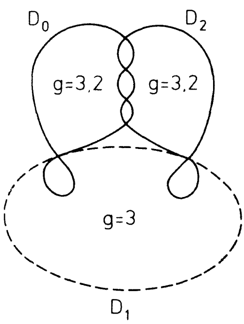
Theorem 4. *Since the Dynkin diagrams for Kac-Moody Lie algebras $\mathscr{a}_2^{(1)}, \mathscr{c}_2^{(1)}, \mathscr{g}_2^{(1)}, \mathscr{a}_4^{(2)}, \mathscr{d}_3^{(2)}, \mathscr{d}_4^{(3)}$ all have three vertices, the Toda equations lead to a divisor of 3 curves (at ∞), completing the affine invariant surfaces into Abelian surfaces. These divisors depicted in Table 2, intersect according to the four patterns listed in Table 1. Each of the genera are given in the picture; the integers appearing after the comma correspond to the genera of the smooth version, if the original one was singular.*

Table 2. The Zoo: curves completing the Toda invariant surfaces to Abelian surfaces. The singularities are those of Table 1



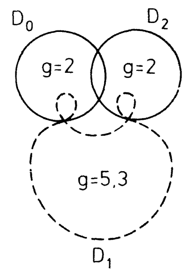
$a_2^{(1)}$	$a_4^{(2)}$
α_0	$p: 1 \quad 2 \quad 2$
$\alpha_1 \quad \alpha_2$	$\cdot \Rightarrow \cdot \Rightarrow \cdot$
	$\hat{p}: 1 \quad 1 \quad 1/2$
	$\alpha_0 \quad \alpha_1 \quad \alpha_2$
$3D_0 \simeq 3D_1 \simeq 3D_2$	$D_1 \simeq D_0$
polar.:	$2D_2 \simeq D_0$
(1, 1)	polar.: (1, 1)

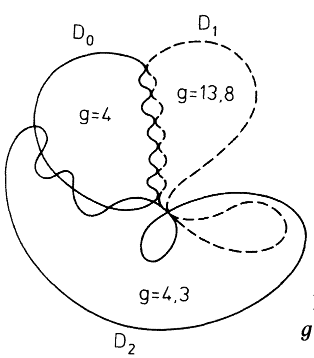




dual

$c_2^{(1)}$	$d_3^{(2)}$
$p: 1 \quad 2 \quad 1$	$1 \quad 1 \quad 1 : p$
$\cdot \Rightarrow \cdot \Leftarrow \cdot$	$\cdot \Leftarrow \cdot \Rightarrow \cdot$
$\hat{p}: 1 \quad 1 \quad 1$	$1 \quad 2 \quad 1 : \hat{p}$
$\alpha_0 \quad \alpha_1 \quad \alpha_2$	$\alpha_0 \quad \alpha_1 \quad \alpha_2$
$2D_1 \simeq 2D_0$	$D_1 \simeq 2D$
$D_2 \simeq D_0$	$2D_2 \simeq 2D_0$
polar.: (1, 2)	polar.: (1, 1)





dual

$g_2^{(1)}$	$d_4^{(3)}$
$1 \quad 2 \quad 3$	$1 \quad 2 \quad 1 : p$
$\cdot - \cdot \Rrightarrow \cdot$	$\cdot - \cdot \Lleftarrow \cdot$
$1 \quad 2 \quad 1$	$1 \quad 2 \quad 3 : \hat{p}$
$\alpha_0 \quad \alpha_1 \quad \alpha_2$	$\alpha_0 \quad \alpha_1 \quad \alpha_2$
$D_1 \simeq 2D_0$	$D_1 \simeq 2D_0$
$D_2 \simeq D_0$	$D_2 \simeq 3D_0$
polar.: (1, 3)	polar.: (1, 1)

g = genus, genus of smooth version.

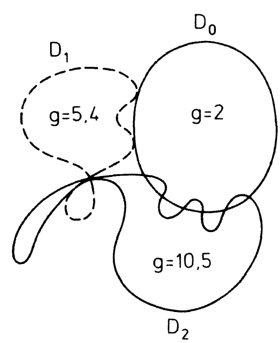


Table 3. 3-Body Toda lattice invariants

$\prod x_i^{p_i}$		
$\delta_0\varphi_0+\delta_2\varphi_2-4\delta_1 x_1$	(except for $a_2^{(1)}$)	$\delta_i=p/\hat{p}_i$ $pA=0,\ A\hat{p}=0$ $\varphi_i=y_i^2-4x_i$
$a_2^{(1)}$	$(y_2-y_0)(y_0-y_1)(y_1-y_2)+9x_0(y_1-y_2)+9x_1(y_2-y_0)+9x_2(y_0-y_1)$	
$a_4^{(2)}$	$\varphi_0\varphi_2-4x_1(y_0y_2-4x_2-x_1)$	
$c_2^{(1)}$	$\varphi_0\varphi_2-8x_1(y_0y_2-2x_1)$	
$d_3^{(2)}$	$\varphi_0\varphi_2-x_1(2y_0y_2-4x_0-x_1-4x_2)$	
$g_2^{(1)}$	$27(\varphi_2-\varphi_0)^2\varphi_2+16x_1[18x_0yy_2+3x_1y^2+6x_2y(3y+y')+2y'y^2y_2-108x_2(x_0+x_2)]$	
where $y=y_1+2y_0,\quad y'=y_1-y_0$.		

§ 2. The intersection of divisors – a Painlevé analysis (local theory)

In this section, we provide the tools needed to prove the results stated in § 1; the proofs will be broken up into several propositions. Since some of the basic facts are known, we merely sketch their proof, with the purpose of making this paper reasonably self-contained.

Proposition 1. *The Hamiltonian system*

$$X_\Delta : \dot{x}_\alpha = x_\alpha y_\alpha, \qquad \alpha\in\Delta, \qquad \dot{y}=A_\Delta x, \tag{2.1}$$

has, besides the Casimir functions $H_0=\prod_0^l x_i^{p_i}$, *l invariants* $H_1,\ldots,H_l$ *in involution. They are extensions of the Weyl polynomials of* Δ_0, *and thus have weighted degrees* $m_1+1, m_2+1,\ldots,m_l+1$ $(m_i=m_i(\Delta_0))$. *The* H_i *generate commuting flows which are straight line motions on Abelian varieties* T^l, *where*

$$T^l\backslash Divisor = A = \bigcap_0^l \{H_i(x,y)=c_i,\ (x,y)\in\mathbb{C}^{2l+1}\}.$$

Proof (sketch). In (1.4), we gave the Lax pair $\dot{L}=[L,P]\in L(g,v)$ for the Toda equations (2.1), with root vectors $e_{\alpha_i}(1\leqq i\leqq l)$ going with the roots α_i of the finite dimensional Lie algebra g; moreover $e_{\alpha_0}=s\bar{e}_{\alpha_0}$, in terms of a root vector $\bar{e}_{\alpha_0}$ in g_0 as well. Thus we have

$$L=L(s)=\sum_1^l y_i\hat{\lambda}_i-\sum_1^l x_ie_{\alpha_i}-sx_0\bar{e}_{\alpha_0}+\sum_0^l e_{-\alpha_i}+s^{-1}\bar{e}_{-\alpha_0},$$

$$P=P(s)=-\sum_1^l x_ie_{\alpha_i}-sx_0\bar{e}_{\alpha_0}.$$

Then, since P and L are polynomials in s and s^{-1}, so is $\varrho(P)$ and $\varrho(L)$ for an arbitrary representation. Then, since P and L satisfy $P-L=O(1)$ when $s\ \infty$, so does $\varrho(P)$ and $\varrho(L)$, and it follows from Theorem 2.6 in [3] that for an arbitrary representation of g, the vector field

$$\varrho(L)^{\boldsymbol{\cdot}}=[\varrho(L),\varrho(P)]$$

linearizes on the Jacobian $\mathscr{J}(\mathscr{C}_\varrho)$ of the (time-independent) spectral curve $\mathscr{C}_\varrho$:

$$\text{affine part of } \mathscr{C}_\varrho : \{(z,s), Q(z,s) \equiv \det(zI-\varrho(L)) = \sum_{i,j} z^i s^j H_{ij}(y,y) = 0\},$$

as do all the Kostant-Kirillov flows generated by the constants of the motion $H_{ij}(x,y)$; indeed, to each H one associates a matrix P_H and there is still – between L and P_H –a relation of the form

$$\varrho(P_H) - \text{polynomial}\,(\varrho(L), s, s^{-1}) = O(1), \text{ when } s \nearrow \infty .$$

The linearizing map to $\mathscr{J}(\mathscr{C}_\varrho)$ is given in terms of a basis of g holomorphic differentials of $\mathscr{C}_\varrho$, by

$$i_\varrho : \mathscr{A} \to \mathscr{J}(\mathscr{C}_\varrho) : \varrho(L) \rightsquigarrow \sum_i \int_{0_i}^{\mu_i} (\omega_1, \dots, \omega_g),$$

where

$$\varrho(L) v_\varrho(s,z) = z v_\varrho(s,z) \quad \text{and} \quad \sum \mu_i = \text{poles of } v_\varrho(s,z) \text{ in the affine part of } \mathscr{C}_\varrho .$$

The map i_ϱ is locally injective if ϱ is faithful (which we assume).

Setting $x_0 = 0$, the polynomials $H_{i0}(x,y)$ are invariant under the coadjoint action of $\mathscr{g}_0$. Therefore by Chevalley's Theorem they are extensions of the Weyl polynomials $H_{i0}(o,y)$ of $\mathscr{g}_0$, for which there exists a basis having degree m_i+1 in the y; therefore the $H_{ij}(x,y)$ possess a basis having weighted degrees $m_1+1, \dots, m_l+1$. To conclude, $i_\varrho(\mathscr{A}) \subset \mathscr{J}(\mathscr{C}_\varrho)$ is an l-dimensional subabelian variety with all its linear flows generated by the invariants H_{ij}. A little arguing shows that not only does $i_\varrho(\mathscr{A})$ complete into an abelian variety $\overline{i_\varrho(\mathscr{A})}$, but $\mathscr{A}$ also completes into an Abelian variety T^l, isogenous to $\overline{i_\varrho(\mathscr{A})}$. This ends the proof of Proposition 1.

Proposition 2. *For each $\Delta' \subset \Delta, 1 \leqq |\Delta'| \leqq l$, there exists Laurent solutions of the system $X_{\Delta'}$ of the form given in Theorem 3 (I); they contain at most $2l+1$-$|\Delta'|$ free parameters, and these are the only Laurent solutions of the vector field X_Δ.*

The proof of Proposition 2 requires the following Lemmas, whose proofs can be found in [1] and [2].

Lemma A. *Consider a set $\Delta = \{\alpha_0, \dots, \alpha_l\}$ of $l+1$ vectors $\alpha_i \in \mathbb{R}^{l+1}$, a non-degenerate real inner product on $\mathbb{R}^{l+1}$, and a matrix*

$$A_\Delta = \left(\frac{2\langle \alpha_i, \alpha_j \rangle}{\langle \alpha_j, \alpha_j \rangle} \right)_{0 \leqq i,j \leqq l} \quad \text{of rank } A_\Delta = l,$$

such that the null-vector p of A_Δ^T (i.e. $A_\Delta^T p = 0$) satisfies

$$(2.2) \qquad \sum p_i \neq 0 \quad \text{and all} \quad p_i \neq 0 .$$

Also consider the differential equations determined by Δ:

$$X_\Delta : \dot{x}_i = x_i y_i, \quad \dot{y} = A_\Delta x \quad x, y \in \mathbb{C}^{l+1} .$$

Then their only formal pole solutions

$$x(t) = \frac{x^{(m)}}{t^m} + \frac{x^{(m-1)}}{t^{m-1}} + \dots, \qquad y(t) = \frac{y^{(k)}}{t^k} + \frac{y^{(k-1)}}{t^{k-1}} + \dots$$

(with m or $k>0$ and $x^{(m)}, y^{(k)} \neq 0$) must match the weights of x and y, i.e. $m=2$ and $k=1$ (Linear convexity argument).

Lemma B. *In a weight-homogeneous system $\dot{z}_i = f_i(z)$, $1 \leqq i \leqq n$, where the z_i's have weight $v_i \in \mathbb{Z}$, formal pole solutions of the form*

$$z_i(t) = \frac{1}{t^{v_i}} (z_i^{(0)} + z_i^{(1)} t + \ldots), \quad 1 \leqq i \leqq n, \quad \textit{some } z_j^{(0)} \neq 0,$$

are convergent Laurent series (majorant method).

Lemma C. *Consider a Lie algebra generated by the simple indecomposable set $\Delta' = \{\alpha_1, \ldots, \alpha_k\}$ of roots and $2\varrho_{\hat{\Delta}'} = \sum_1^k \hat{n}_{\alpha_i} \hat{\alpha}_i$. Then the operator*

$$\textit{diagonal}\,(\hat{n}_{\alpha_1}, \ldots, \hat{n}_{\alpha_k}) \cdot A_{\Delta'},$$

has for spectrum

$$m_1(m_1+1), m_2(m_2+1), \ldots, m_k(m_k+1),$$

where $m_1, \ldots, m_k$ are the Weyl exponents of that Lie algebra generated by Δ'.

Proof of Proposition 2 (sketch). We first remark that (2.2) holds for all Cartan matrices A_Δ. By Lemma A, we know that the only possible formal pole solutions of X_Δ are of the form

$$(2.3) \qquad x(t) = \frac{1}{t^2} (x^0 + x^1 t + \ldots), \quad y(t) = \frac{1}{t} (y^0 + y^1 t + \ldots), \quad x^0 \text{ and } y^0 \neq 0.$$

By Lemma B, such formal solutions are automatically convergent, and so it suffices to look for formal solutions of the form (2.3). Some x_i^0 must vanish; indeed if all $x_i^0 \neq 0$, then all $y_i^0 = -2$; since $\sum p_i y_i$ is invariant under the flow X_Δ, we have

$$0 = \sum p_i y_i^0 = -2 \sum p_i,$$

a contradiction of (2.2). Then, after possibly relabeling, we assume the first s lead terms $\neq 0$ and the remaining $= 0$, with $0 \leqq s \leqq l-1$. For $z \in \mathbb{C}^{l+1}$, introduce the notation

$$\bar{z} = (z_0, z_1, \ldots, z_{s-1}), \quad \underline{z} = (z_s, \ldots, z_l) \quad \text{and} \quad \bar{A} = (A_{ij})_{0 \leqq i,j \leqq s-1}.$$

Then substitute (2.3) into the system X_Δ and, after a little manipulation, equate the coefficients[8] of t^{k-3}, keeping in mind the relation $\sum_0^l p_i y_i^k = 0$:

$$(2.4) \qquad k=0 \begin{cases} \bar{y}^0 + 2\bar{\delta} = 0 \\ y^0 = -Ax^0 \\ \bar{A}\bar{x}^0 = 2\bar{\delta}; \quad \underline{x}^0 = 0 \end{cases}$$

$$k=1 \begin{cases} \bar{y}^1 = (\bar{x}^0)^{-1} * \bar{x}^1 \\ Ax^1 = 0 \\ \underline{x}^1 * (\underline{y}^0 + \underline{\delta}) = 0 \end{cases}$$

$$k \geqq 2 \begin{cases} [\bar{x}^0 * \bar{A} - k(k-1)]\bar{x}^k = (k-1)\bar{\Gamma}_{k-1} \\ (k-1)y^{(k)} = Ax^{(k)} \\ \underline{x}^k * (\underline{y}^0 - (k-2)\underline{\delta}) = \underline{\Gamma}_{k-1} \end{cases}$$

[8] $\delta = (1, 1, 1, \ldots, 1)$ and for $a, b \in \mathbb{C}^n$, define $a * b = (a_1 b_1, \ldots, a_n b_n) \in \mathbb{C}^n$

where

$$\Gamma_{k-1} \equiv \Gamma_{k-1}(x^0,\dots,x^{k-1},y^0,\dots,y^{k-1}) = -\sum_{j=1}^{k-2} \frac{x^j * Ax^{k-j}}{k-j-1} - x^{k-1} * y^1 . \tag{2.5}$$

At first we have $x^1=0$; indeed since $Ax^1=0$ (see $k=1$), we have $x^1=c\hat{p}$ with all $\hat{p}_i \neq 0$ by (2.2), and so all x_i would have a pole if $c \neq 0$ were to hold; this is absurd, since $H_0=\prod x_i^{p_i}=1$. Looking at the relations for $k=0$, i.e. $\underline{x}^0=0$ and $\bar{A}\bar{x}^0=2\bar{\delta}$, amounts to

$$(\bar{A}\bar{x}^0)_i = \left\langle \alpha_i, \sum_0^{s-1} x_j^0 \hat{\alpha}_j \right\rangle = 2 ;$$

according to (1.12), this yields

$$\sum_0^{s-1} x_j^0 \hat{\alpha}_j = 2\varrho_{\hat{\Delta}'} = \sum \hat{n}_j \hat{\alpha}_j$$

in terms of the element 2ϱ associated with $\hat{\Delta}=\{\hat{\alpha}_0,\hat{\alpha}_1,\dots,\hat{\alpha}_{s-1}\}$. Thus we have

$$\begin{aligned} x_j^0 &= \hat{n}_j \quad \text{for} \quad 0 \leqq j \leqq s-1 \\ &= 0 \quad \text{for} \quad s \leqq j \leqq l . \end{aligned} \tag{2.6}$$

Now we use the $(k=0)$-relation $y^0=-Ax^0$

$$\begin{aligned} y_i^0 = -(Ax^0)_i &= -\left\langle \alpha_i, \sum_0^{s-1} x_j^0 \hat{\alpha}_j \right\rangle = -\langle \alpha_i, 2\varrho_{\hat{\Delta}'} \rangle = -2 , \\ &\qquad (\text{for } 0 \leqq i \leqq s-1) \\ &= -\sum_{j=0}^{s-1} A_{ij}\hat{n}_j, \text{ in general}. \end{aligned} \tag{2.7}$$

Since $x^1=0$ and since there is no constraint on y^1 except for $\sum p_i y_i^1=0$ and the $(k=1)$-relation $\bar{y}^1=(\bar{x}^0)^{-1} * \bar{x}^{(1)}$, we have

$$y^1=(0,\dots,0,c_s,\dots,c_l), \quad \sum p_k c_k = 0 , \tag{2.8}$$

leading to $l-s$ free parameters.

Moving on to $k \geqq 2$, we have from (2.5), (2.7) and (2.8) the relation

$$x_j^k(2-\langle \alpha_i, 2\varrho_{\hat{\Delta}'}\rangle - k) = -\sum_{r=1}^{k-2} \frac{x_j^r (Ax^{k-r})_j}{k-r-1} - x_j^{k-1} c_j, \qquad \text{for } x \leqq j \leqq l$$

which, by induction, yields

$$\left.\begin{aligned} x_j^k &= 0 \quad \text{for} \quad k < 2 - \langle \alpha_i, 2\varrho_{\hat{\Delta}'} \rangle \\ &= \beta_j (\text{free parameter}) \text{ for } k = 2 - \langle \alpha_i, 2\varrho_{\hat{\Delta}'} \rangle \end{aligned}\right\} s \leqq j \leqq l .$$

Then, using Lemma C and (2.6), the operator $\bar{x}^0 * \bar{A}_{\Delta'}$ has for spectrum the integers $m_1(m_1+1),\dots,m_k(m_k+1)$, all $\geqq 2$ with possible repetitions when Δ' is decomposable; the m_i are the Weyl exponents of the Lie algebra generated by Δ'. Thus for $k \neq m_i+1$,

$$\bar{x}^{(k)} = [\bar{x}^{(0)} * \bar{A} - k(k-1)]^{-1}(k-1)\bar{\Gamma}_{k-1} ,$$

while for $k=m_i+1$, the equation for $\bar{x}^k$ is solvable with the number of parameters equal to the multiplicity of $k(k-1)$, provided the right hand side $(k-1)\bar{\Gamma}_{k-1}$ of the linear system belongs to the image of the left hand side. Thus at most $|\Delta'|$ free parameters enter the solutions here, and exactly $|\Delta'|$ only if the solvability conditions are met. This concludes the proof of Proposition 2, by observing that $(|\Delta'|=s)$,

$$\#\begin{pmatrix}\text{definite}+\\ \text{possible parameters}\end{pmatrix}=[(l-s)+(l-s+1)]+|\Delta'|=2l+1-|\Delta'|,$$

$$(\text{since } s=|\Delta'|)$$

occur in the series.

Proof of Theorem 3 (sketch).

Part I. In Proposition 2, we have shown that the Laurent solutions $D_{\Delta'}(t)$ contain *at most* $2l+1-|\Delta'|$ free parameters; in fact one shows in a separate argument, which we omit, that *all* parameters actually occur. The statement concerning the weighting of $x(t,b,c,d)$ and $y(t,b,c,d)$ follows from the formulas (2.4), which show (inductively) that the weight of x^k and y^k is k.

To prove the last statement of Part I, observe by (1.34) that

$$x_{\beta|D_{\{\alpha\}}(t)}=b_\beta t^{-A_{\beta\alpha}}+\dots; \tag{2.9}$$

so for $\Delta'\subset\Delta$, if

$$2\varrho_{\Delta'}=\sum_{\Delta'} n_\beta\beta \quad\text{and}\quad |2\varrho_{\Delta'}|=\sum_{\Delta'} n_\beta,$$

we have the following, by (2.9)

$$\prod_{\beta\in\Delta'} x_\beta^{-n_\beta/2}|_{D_{\{\alpha\}}(t)}=\begin{pmatrix}\text{expression}\\ \neq 0\end{pmatrix}\cdot t^{1/2\sum_\beta n_\beta A_{\beta\alpha}}.$$

Since $\langle\varrho_{\Delta'},\hat{\alpha}\rangle=1$ for all $\alpha\in\Delta'$, we also have

$$\frac{1}{2}\sum_{\beta\in\Delta'} n_\beta A_{\beta\alpha}=\frac{1}{2}\sum_{\beta\in\Delta'} n_\beta\langle\beta,\hat{\alpha}\rangle=\langle\varrho_{\Delta'},\hat{\alpha}\rangle=1,$$

and so

$$\prod_{\beta\in\Delta'} x_\beta^{-n_\beta/2}|_{D_{\{\alpha\}}(t)}=O(t),\qquad \alpha\in\Delta',$$

as claimed. Finally for the solution (1.28)

$$D_{\Delta'}(t): x_\beta=\hat{n}_\beta t^{-2}+\dots,\qquad \beta\in\Delta',$$

compute

$$\prod_{\beta\in\Delta'} x_\beta^{-n_\beta/2}=\left(\prod_{\beta\in\Delta'}\hat{n}_\beta^{-n_\beta/2}\right)t^{\Sigma n_\beta}=\left(\prod_{\beta\in\Delta'}\hat{n}_\beta^{-n_\beta/2}\right)t^{|2\varrho_{\Delta'}|},$$

confirming (1.29). This ends the proof of Part I.

Part II. Since the Toda problem is algebraic integrable, Theorem 1 of [2] implies the existence of a coherent tree of Laurent solutions, satisfying all the conditions described in [2]. Since we have shown that the $D_{\Delta'}(t)$ are all the Laurent solutions of the Toda system, it suffices to show they fit together as stated in Theorem 3. We now briefly explain this concept: each of the Laurent solutions $D_{\Delta'}(t)$, $\phi \neq \Delta' \subsetneqq \Delta$ forms a fibre bundle over $D_{\Delta'}$, which can be assembled into a *coherent tree*, yielding a fibre bundle over $D = \cup D_\alpha$. A *coherent tree* is a partially ordered set of affine varieties: the inequality $D_{\Delta''} < D_{\Delta'}$ between adjacent varieties means that dimension$(D_{\Delta''})$ = dimension$(D_{\Delta'}) - 1$ and that the fibre bundle $D_{\Delta''}(t)$ can be *glued* to $D_{\Delta'}(t)$. These inequalities concatenate then into a partial ordering. If $D_{\Delta''}(t)$ is glued to $D_{\Delta_1}(t)$ and $D_{\Delta_2}(t)$, then D_{Δ_1} and D_{Δ_2} are sewn together along $D_{\Delta''}$.

Checking Fig. 1 amounts to verifying the existence of a typical subtree (let $\Delta' = \{\alpha_0, \dots, \alpha_s\}$)

$$\begin{tikzcd}
D_{\alpha_0} \arrow[drr, dashed] & D_{\alpha_1} \arrow[dr, dashed] & & D_{\alpha_2} \arrow[dl, dashed] \quad \cdots & D_{\alpha_s} \arrow[dll, dashed] \\
& & D_{\Delta'} & &
\end{tikzcd}$$

i.e. showing the existence of a birational transformation

$$T: (x, y) \rightsquigarrow (z_1, \dots, z_N)$$

having the property that

$$T(x, y)|_{D_{\{\alpha_i\}}(t),\, 0 \leqq i \leqq s} = (z_1(t, u), \dots, z_N(t, u))$$

is a vector of holomorphic series in (t, u), $|t|$, $|u| < \varepsilon$, where u accounts for the free parameters. Then

$$(2.7) \qquad T^{-1}(z_1(t, u), \dots, z_N(t, u)) = (\tilde{x}(t, u), \tilde{y}(t, u)) \quad (0 \leqq i \leqq s)$$

$$= \left(\left(\dots, \frac{f_j(t, u)}{g_j(t, u)}, \dots \right), \left(\dots, \frac{f'_j(t, u)}{g'_j(t, u)}, \dots \right) \right)$$

are meromorphic extensions of $(x(t, u), y(t, u))$. Then one checks, by Picard's theorem on the uniqueness of solutions of holomorphic differential equations, that the vector (2.7) of Laurent solutions in t tends to the same $D_{\Delta'}(t)$, whatever be the Laurent solution $D_{\{\alpha_i\}}(t)$, $0 \leqq i \leqq s$, after letting $u \to u_0$ and then dividing the t-series obtained. One checks that in our case, T is given by (1.32), (1.33) and $y_\alpha = \dot{x}_\alpha / x_\alpha$. "*Local completeness*" means that all branches $D_{\alpha_0}, \dots, D_{\alpha_s}$ going through $D_{\Delta'}$ have been accounted for; this is done by checking that for $\alpha \in \Delta'$:

$$\sum_{\alpha \in \Delta'} (\text{order of pole of } x_i \text{ along } D_{\{\alpha\}}(t)) = \text{multiplicity of } g_i(t, u).$$

That $\dim(D_{\Delta'}) = l - |\Delta'|$ follows from the number of free parameters, $2l + 1 - |\Delta'|$, appearing in $D_{\Delta'}(t)$ and that there are $l + 1$ constants. Finally one verifies *global completeness*, namely that the Laurent solutions sweep out the full image of D in

$\mathbb{P}^{2l+1}_{1,2}$, which amounts to checking

$$\sum_{\phi \neq \Delta' \subsetneq \Delta} d_{\Delta'} = \frac{\prod_0^l (\text{degree of } H_i)\,(\text{l.c.m.}(\text{degrees of } x_i \text{ and } y_i))^{l-1}}{\prod_0^l (\text{degrees of } x_i) \prod_0^l (\text{degrees of } y_i)}$$

$$= \frac{\prod_i (m_i+1)\cdot 2\Sigma p_i \cdot 2^{l-1}}{2^{l+1}}$$

$$= \frac{1}{2}\,|W_{\Delta_0}| \sum_0^l p_i, \qquad \text{using (1.13)},$$

where

$$d_{\Delta'} = \text{degree}\left\{\begin{array}{c} l\text{-1-dimensional components of the image of} \\ (\tilde{x}(o,u), \tilde{y}(o,u)) \text{ in weighted projective space} \\ \mathbb{P}^{2l+1}_{1,2} \equiv \{(x,y,z) \sim (s^2x, sy, sz)\} \end{array}\right\}$$

This completes the proof of Theorem 3.

The next theorem shows the behavior of the Laurent solutions $D_{\{\alpha\}}(t)$ as they approach $D_\alpha \cap D_\beta$. The coefficients in the series $D_{\{\alpha\}}(t)$ will be expressed in terms of interesting holomorphic local coordinates on T^l defined by functions and their derivatives.

Proposition 3. *(Local behavior of $D_{\{\alpha\}}(t)$ near $D_{\alpha\beta}(t)$ in terms of local parameters on T^l). The free parameters b and c in the Laurent solution*

(2.10)

$$D_{\{\alpha\}}(t): \begin{cases} x_\alpha = t^{-2} + b_\gamma + \dots \\ x_\gamma = b_\gamma t^{-A_{\gamma\alpha}}(1 + c_\gamma t + \dots) \end{cases}, \quad \begin{cases} y_\alpha = -2t^{-1} + 2b_\alpha t + \dots \\ y_\gamma = -A_{\gamma\alpha} t^{-1} + c_\gamma + \dots \end{cases}, \quad \sum_{\gamma \neq \alpha} p_\gamma c_\gamma = 0$$

approach the parameters b' and c' in the Laurent solution

$$(2.11) \qquad D_{\alpha\beta}(t): \begin{cases} x_\alpha = \hat{n}_\alpha t^{-2} + O(b'_\alpha, b'_\beta) \\ x_\beta = \hat{n}_\beta t^{-2} + O(b'_\alpha, b'_\beta) \\ x_\gamma = b'_\gamma t^{-(\hat{n}_\gamma A_{\gamma\alpha} + \hat{n}_\gamma A_{\gamma\beta})}(1 + c'_\gamma t + \dots) \quad \gamma \neq \alpha, \beta \end{cases}$$

$$\left(\sum_{\gamma \neq \alpha,\beta} p_\gamma c'_\gamma = 0\right)$$

$$\begin{cases} y_\alpha = -2t^{-1} + O(t) \\ y_\beta = -2t^{-1} + O(t) \\ y_\gamma = -(\hat{n}_\alpha A_{\gamma\alpha} + \hat{n}_\beta A_{\gamma\beta})t^{-1} + c'_\gamma + \dots, \quad \gamma \neq \alpha, \beta \end{cases}$$

in the way described below. We distinguish four typical situations for the solution $D_{\alpha\beta}(t)$. All the finite parameters below are local parameters on T^l, with $D_\alpha \cap D_\beta \subset \{s=0\}$.

Case (i) *when* $A_{\alpha\beta}=0$, $\quad \cdots\ \underset{\gamma}{\cdot} \quad \underset{\alpha}{-} \quad \cdots \underset{\beta'}{\bullet}\!-\!\underset{\beta}{\bullet}\!-\!\underset{\beta''}{\bullet}$

$$b_\gamma^{(2)}\to b'_\gamma, \qquad c_\gamma^{(1)}\to c'_\gamma, \qquad \gamma\neq\beta,\beta',\beta''$$ [9]

$$b_\beta^{(2)}=1/4s^2+\ldots, \qquad c_\beta^{(1)}=1/s, \qquad b_\alpha^{(2)}=O'(1)$$

$$b_\gamma^{(3)}=O'(s), \qquad c_\gamma^{(1)}=O'(1)/s, \qquad \gamma=\beta' \text{ or } \beta''.$$ [10]

Case (ii) *when* $A_{\alpha\beta}=-1$, $\quad \underset{\gamma}{\bullet}\ \cdots\ \underset{\alpha'}{\bullet}\!-\!\underset{\alpha}{\bullet}\!-\!\underset{\beta}{\bullet}\!-\!\underset{\beta''}{\bullet}\ \cdots\ \underset{\gamma}{\bullet}$

$$b_\gamma^{(2)}\to b'_\gamma, \qquad c_\gamma^{(1)}\to c'_\gamma, \qquad \gamma\neq\alpha,\alpha',\beta,\beta'$$

$$b_\beta^{(3)}=(-2/3s)^3+\ldots, \quad c_\beta^{(1)}=1/s, \qquad b_\alpha^{(2)}=1/(3s)^2+\ldots$$

$$b_{\alpha'}=O'(s), \qquad c_{\alpha'}^{(1)}=O'(1)/s,$$

$$b_{\beta'}=O'(s^2), \qquad c_{\beta'}^{(1)}=O'(1)/s.$$

Case (iii). (a) *when* $A_{\alpha\beta}=-2$ *and* $A_{\beta\alpha}=-1$, $\quad \cdots\ \underset{\alpha}{\bullet}\Rightarrow\underset{\beta}{\bullet}\!-\!\underset{\beta'}{\bullet}\ \cdots\ \underset{\gamma}{\bullet}\ \cdots$

the solution $D_{\{\alpha\}}(t)$ *tends to* $D_{\alpha\beta}(t)$ *in two different ways:*

$$b_\gamma^{(2)}\to b'_\gamma, \qquad c_\gamma^{(1)}\to c'_\gamma, \qquad \gamma\neq\alpha,\beta,\beta'$$

$$1)\ b_\beta^{(3)}=(-8/s)^3+\ldots, \qquad c_\beta^{(1)}\equiv\frac{1}{s}, \qquad b_\alpha^{(2)}=\frac{1}{3s^2}+\ldots,$$

$$b_{\beta'}^{(2)}=O'(s^3), \qquad c_{\beta'}^{(1)}=-c_\beta^{(1)}+O'(1)$$

or

$$2)\ b_\beta^{(3)}=\frac{O'(1)}{s}, \qquad c_\beta^{(1)}\equiv s, \qquad b_\alpha^{(2)}=O'(1);$$

$$b_{\beta'}^{(2)}=O'(s), \qquad c_{\beta'}^{(1)}=O'(s).$$

(b) *when* $A_{\alpha\beta}=-1$ *and* $A_{\beta\alpha}=-2$, $\quad \cdots\ \underset{\beta}{\bullet}\Rightarrow\underset{\alpha}{\bullet}\!-\!\underset{\alpha'}{\bullet}\ \cdots\ \underset{\gamma}{\bullet}\ \cdots$

the solution $D_{\{\alpha\}}(t)$ *tends to* $D_{\alpha\beta}(t)$ *as follows:*

$$b_\gamma^{(4)}\to b'_\gamma, \qquad c_\gamma^{(1)}\to c'_\gamma, \qquad \gamma\neq\alpha,\alpha',\beta,$$

$$b_\beta^{(1)}=\frac{1}{4s^4}+\ldots, \qquad c_\beta^{(1)}\equiv\frac{1}{s}, \qquad b_\alpha^{(2)}=\frac{1}{12s^2}+\ldots,$$

$$b_{\alpha'}^{(3)}=O'(s^2), \qquad c_{\alpha'}^{(1)}=-c_\beta^{(1)}+O'(1).$$

Case (iv). (a) *When* $A_{\alpha\beta}=-3$ *and* $A_{\beta\alpha}=-1$, $\quad \underset{\gamma}{\bullet}\!-\!\underset{\alpha}{\bullet}\Rrightarrow\underset{\beta}{\bullet}$

the solution $D_{\{\alpha\}}(t)$ *tends to* $D_{\alpha\beta}(t)$ *as follows:*

[9] the superscripts in the b and c refer to the weighted degrees in the sense of Theorem 3
[10] there is a b_α, but no c_α. O' denotes growth in s with dependence on b' and c'

$$b^{(3)}_\beta = \frac{(8\pm 16/\sqrt{3})}{s^3} + \ldots, \qquad c^{(1)}_\beta \equiv \frac{1}{s}, \qquad b^{(2)}_\alpha = \frac{1}{s^2} + \ldots,$$
$$b^{(3)}_\gamma = O(s^9), \qquad c^{(1)}_{\alpha'} = -3c^{(1)}_\beta$$

(b) *when* $A_{\alpha\beta} = -1$ *and* $A_{\beta\alpha} = -3$, $\underset{\gamma\quad\beta}{\bullet\!\!-\!\!\!-\!\!\!-\!\!\bullet} \Rrightarrow \underset{\alpha}{\bullet}$

the solution $D_{\{\alpha\}}(t)$ *tends to* $D_{\alpha\beta}(t)$ *in two distinct ways:*

1) $b^{(5)}_\beta = 8/27s^5 + \ldots, \quad c^{(1)}_\beta \equiv \frac{1}{s}, \quad b^{(2)}_\alpha = \frac{1}{(3s)^2} + \ldots,$
$b^{(2)}_\gamma = O(s^{10}), \quad c^{(1)}_\gamma = -2c^{(1)}_\beta$

2) $b^{(5)}_\beta = O'(1)/s, \quad c^{(1)}_\beta \equiv s, \quad b^{(2)}_\alpha = O(1),$
$b^{(2)}_\gamma = O(s^2), \quad c^{(1)}_\gamma = -2c^{(1)}_\beta.$

To sum up, in all but the cases denoted by 2), the parameters $b^{(2)}_\alpha, b^{(k)}_\beta$ *and* $c^{(1)}_\beta$ *blow up according to their weights* $2, k$ *and* 1 *respectively (except for* $b^{(2)}_\alpha$ *in case* (i)*); however in case 2), there is a deficiency: we have* $c^{(1)}_\beta = s, b^{(2)}_\alpha = O'(1)$ *and* $b^{(k)}_\beta = O'(1)/s$ *with* $k=3$ *and* 5 *in cases (iii) and (iv) respectively. Some of the "neighboring" parameters* $c^{(1)}_\gamma$ *may blow up, but never with more than a simple pole, while the more "distant" parameters approach the parameter of the* $D_{\alpha\beta}(t)$ *solution. A similar state of affairs holds for the other cases of* $\Delta' = \{\alpha, \beta\}$.

Proof. Case (i). Since in this case $2\varrho_{\{\hat\alpha,\hat\beta\}} = \hat\alpha + \hat\beta$, the Laurent solutions $D_{\{\alpha\}}(t)$, $D_{\{\beta\}}(t)$ and $D_{\alpha\beta}(t)$ become[11]

$$\begin{array}{lll}
D_{\{\alpha\}}(t): & x_\alpha = t^{-2} + b_\alpha + \ldots, & x_\alpha^{-1} = t^2 - t^4 b_\alpha + \ldots \\
& x_\beta = b_\beta(1 + c_\beta t + \ldots) & (x_\alpha x_\beta)^{-1/2} = (b_\beta)^{-1/2} t(1 - c_\beta t/2 + \ldots) \\
& x_\gamma = b_\gamma t^{-A_{\gamma\alpha}}(1 + c_\gamma t + \ldots), & \gamma \neq \alpha, \beta \\
D_{\{\beta\}}(t): & x_\alpha = \tilde b_\alpha(1 + \tilde c_\alpha t + \ldots), & x_\alpha^{-1} = (\tilde b_\alpha)^{-1}(1 - \tilde c_\alpha t + \ldots) \\
& x_\beta = t^{-2} + \tilde b_\beta + \ldots & (x_\alpha x_\beta)^{-1/2} = (\tilde b_\alpha)^{-1/2} t(1 - \tilde c_\alpha t/2 + \ldots) \\
& x_\gamma = \tilde b_\gamma t^{-A_{\gamma\beta}}(1 + \tilde c_\gamma t + \ldots), & \gamma \neq \alpha, \beta \\
D_{\alpha\beta}(t): & x_\alpha = t^{-2} + b'_\alpha + \ldots, & x_\alpha^{-1} = t^2 - t^4 b'_\alpha + \ldots \\
& x_\beta = t^{-2} + b'_\beta + \ldots & (x_\alpha x_\beta)^{-1/2} = t^2 t(1 - t^2(b'_\alpha + b'_\beta)/2 + \ldots \\
& x_\gamma = b'_\gamma t^{-(A_{\gamma\alpha} + A_{\gamma\beta})}(1 + c'_\gamma t + \ldots), & \gamma \neq \alpha, \beta.
\end{array}$$

In a neighborhood of $D_\alpha \cap D_\beta$, the functions $x_\alpha^{-1}, (x_\alpha x_\beta)^{-1/2}$ and $x_\gamma(\gamma \neq \alpha, \beta)$ are finite, since they are finite in the affine, and since no other $D_\gamma(\gamma \neq \alpha, \beta)$ contains a component of the intersection $D_\alpha \cap D_\beta$ (Theorem 3, II). Therefore, by Hartog's lemma, they are holomorphic about $D_\alpha \cap D_\beta$, and so, the functions $x_\alpha^{-1}, (x_\alpha x_\beta)^{-1/2}$ and x_γ evaluated along the trajectories $D_{\alpha\beta}(t)$ are *bona fide* limits of the analogous series for $D_{\{\alpha\}}(t)$ (or $D_{\{\beta\}}(t)$). This yields at once the term by term estimates for the

[11] besides $g_{\{\alpha\beta\}} \equiv (x_\alpha x_\beta)^{-1/2}$ of Theorem 3.I, we need more holomorphic functions

local behavior of the free parameters

$$(2.12)\qquad \begin{array}{lll} b_\gamma \to b'_\gamma, & c_\gamma \to c'_\gamma, & \gamma \neq \alpha, \beta, \beta', \beta'' \\ b_\gamma \to 0, & b_\gamma c_\gamma \to b'_\gamma, & \gamma = \beta' \text{ or } \beta'' \\ b_\beta^{-1/2} \to 0, & b_\beta^{-1/2} c_\beta \to -2 \text{ and } & b_\alpha \to b'_\alpha; \end{array}$$

setting $c_\beta = 1/s$, we get the two first lines of estimates of the claim (i). Using the relation $\prod x_\alpha^{p_\alpha} = 1$, with typically $p_{\beta'} = p_\beta = p_{\beta''}$, one concludes

$$b_\beta b_{\beta'} b_{\beta''} = O'(1).$$

Since $b_\beta = 1/4s^2 + \dots$ it follows that $b_{\beta'} b_{\beta''} = O'(s^2)$ and from the second line of (2.12) conclude

$$(2.13)\qquad c_{\beta'} c_{\beta''} = O'(1)/s^2.$$

Substituting the expansions (2.10) and the estimates above (lines 1 and 2 of (i)) into the Hamiltonian (1.5) yields

$$(2.13)\qquad \begin{aligned} &\frac{1}{2}\langle \lambda_{\beta'}, \lambda_{\beta'} \rangle c_{\beta'}^2 + \frac{1}{2}\langle \lambda_{\beta''}, \lambda_{\beta''} \rangle c_{\beta''}^2 + \langle \lambda_{\beta'}, \lambda_{\beta''} \rangle c_{\beta'} c_{\beta''} \\ &+ \frac{1}{s}[\langle \lambda_\beta, \lambda_{\beta'} \rangle c_{\beta'} + \langle \lambda_\beta, \lambda_{\beta''} \rangle c_{\beta''}] \\ &= \frac{1}{2s^2}(\langle \beta, \beta \rangle^{-1} - \langle \lambda_\beta, \lambda_\beta \rangle) + \dots, \end{aligned}$$

which together with (2.13) yields, upon rescaling variables, the relations

$$(sc_{\beta'}) = O'(1), \qquad (sc_{\beta''}) = O'(1).$$

Thus from the second line of (2.12) deduce

$$b_{\beta'} = O'(s), \qquad b_{\beta''} = O'(s),$$

and the third line of (i), completing the proof of (i).

Case (ii). Using $2\varrho_{\{\hat\alpha, \hat\beta\}} = 2\hat\alpha + 2\hat\beta$, the Laurent solutions $D_{\{\alpha\}}(t)$ and $D_{\alpha\beta}(t)$ become $D_{\{\alpha\}}(t)$:

$$\begin{array}{ll} x_\alpha = t^{-2} + b_\alpha - \frac{1}{4}(b_{\alpha'} + b_\beta)t + \dots, & y_\alpha = -2t^{-1} + 2b_\alpha t - \frac{3}{4}(b_{\alpha'} + b_\beta)t^2 + \dots \\ x_\beta = b_\beta t(1 + c_\beta t + d_\beta t^2 + e_\beta t^3 + \dots), & y_\beta = t^{-1} + c_\beta + \bar d_\beta t + \bar c_\beta t^2 + \dots \\ x_{\beta'} = b_{\beta'}(1 + c_{\beta'} t & y_{\beta'} = c_{\beta'} + (2b_{\beta'} - b_{\beta''})t + \dots \\ \quad + \frac{1}{2}(c_{\beta'}^2 + 2b_{\beta'} - b_{\beta''})t^2 + \dots), & \end{array}$$

$$(x_\alpha x_\beta)^{-1} = b_\beta^{-1} t(1 - c_\beta t + \tfrac{1}{2}(c_\beta^2 - b_\alpha + b_{\beta'})t^2 + f_\beta t^3 + \dots)$$

$$(x_\alpha^2 x_\beta)^{-1} = b_\beta^{-1} t^3(1 - c_\beta t + \tfrac{1}{2}(c_\beta^2 - 3b_\alpha + b_{\beta'})t^2 + g_\beta t^3 + \dots)$$

$$\begin{aligned} D_{\alpha\beta}(t): x_\alpha &= 2t^{-2} + O(b'_\alpha, b'_\beta) \\ x_\gamma &= b_\gamma t^{-2(A_{\gamma\alpha} + A_{\gamma\beta})}(1 + tc'_\gamma + \dots) \\ x_\beta &= 2t^{-2} + O(b'_\alpha, b'_\beta) \\ (x_\alpha x_\beta)^{-1} &= \tfrac{1}{4} t^{4}(1 + O(t^2)) \\ (x_\alpha^2 x_\beta)^{-1} &= \tfrac{1}{8} t^6(1 + O(t^2)) \end{aligned}$$

with

$$2d_\beta=c_\beta^2-b_\alpha-b_{\beta'},\qquad \bar d_\beta=-b_\alpha-b_{\beta'}$$
$$6e_\beta=2\bar e_\beta+c_\beta(c_\beta^2-3b_\alpha-3b_{\beta'}),\quad 8\bar e_\beta=b_{\alpha'}+9b_\beta-4c_{\beta'}b_{\beta'}.$$

(2.14)
$$f_\beta=\left(-\frac{c_\beta^3}{6}+\frac{b_\alpha c_\beta}{2}-\frac{b_\beta}{8}\right)+\left(\frac{c_{\beta'}b_{\beta'}}{6}+\frac{5b_{\alpha'}}{24}-\frac{b_{\beta'}c_\beta}{2}\right)$$
$$g_\beta=\left(-\frac{c_\beta^3}{6}+\frac{3b_\alpha c_\beta}{2}+\frac{b_\beta}{8}\right)+\left(\frac{b_{\beta'}c_{\beta'}}{6}+\frac{11b_{\alpha'}}{24}-\frac{b_{\beta'}c_\beta}{2}\right).$$

Arguing as before, the functions $x_\gamma(\gamma\neq\alpha,\beta),(x_\alpha x_\beta)^{-1}$ and $(x_\alpha^2x_\beta)^{-1}$ are holomorphic near $D_\alpha\cap D_\beta$, therefore the series x_γ, $(x_\alpha x_\beta)^{-1}$ and $(x_\alpha^2 x_\beta)^{-1}$ along the orbits $D_{\alpha\beta}(t)$ are *bona fide* limits of the analogous series for $D_{\{\alpha\}}(t)$ (or $D_{\{\beta\}}(t)$), yielding term by term the limits

(2.15)
$$\begin{array}{llll}
x_\gamma: b_\gamma\to b_\beta', & c_\gamma\to c_\gamma' & \gamma\neq\alpha,\beta,\alpha',\beta' & \\
x_{\alpha'}: b_{\alpha'}\to 0, & b_{\alpha'}c_{\alpha'}\to b_{\alpha'}' & & \\
x_{\beta'}: b_{\beta'}\to 0, & b_{\beta'}c_{\beta'}\to 0, & \tfrac12 b_{\beta'}(c_{\beta'}^2+2b_{\beta'}-b_{\beta''})\to b_{\beta'}' & \\
(x_\alpha x_\beta)^{-1}: b_\beta^{-1}\to 0, & b_\beta^{-1}c_\beta\to 0, & (2b_\beta)^{-1}(c_\beta^2-b_\alpha+b_{\beta'})\to 0, & b_\beta^{-1}f_\beta\to\tfrac14 \\
(x_\alpha^2x_\beta)^{-1}: b_\beta^{-1}\to 0, & -b_\beta^{-1}c_\beta\to 0, & \tfrac12 b_\beta^{-1}(c_\beta^2-3b_\alpha+b_{\beta'})\to 0, & b_\beta^{-1}g_\beta\to\tfrac18.
\end{array}$$

From (2.15) it follows that $b_{\alpha'},b_{\beta'},b_{\beta'}c_{\beta'},b_\beta^{-1},b_\beta^{-1}c_\beta\to 0$, and so the latter part of f_β/b_β and g_β/b_β in (2.14) tends to zero. At the same time, $f_\beta/b_\beta\to\frac14$, and $g_\beta/b_\beta\to\frac18$, yielding

$$\frac{1}{b_\beta}\left(-\frac{c_\beta^3}{6}+\frac{b_\alpha c_\beta}{2}\right)\to\frac38\quad\text{and}\quad\frac{1}{b_\beta}\left(-\frac{c_\beta^3}{6}+\frac{3b_\alpha c_\beta}{2}\right)\to 0,$$

hence

$$\frac{b_\alpha c_\beta}{b_\beta}\to-\frac38,\quad\frac{c_\beta^3}{b_\beta}\to-\left(\frac32\right)^3.$$

Thus setting

$$c_\beta\equiv 1/s,$$

one finds

(2.16)
$$b_\beta=\left(-\frac{2}{3s}\right)^3+\ldots,\quad\text{and}\quad b_\alpha=\frac{1}{(3s)^2}+\ldots$$

The estimates of the parameters appearing in $x_{\alpha'}$ and $x_{\beta'}$ (see (2.15)) imply

(2.17)
$$b_{\alpha'}c_{\alpha'}\to b_{\alpha'}'\quad\text{and}\quad b_{\beta'}c_{\beta'}^2\to 2b_{\beta'}'.$$

From the invariant $\prod x_\alpha^{p_\alpha}=1$ (remember generically $\underset{\alpha'}{p}=\underset{\beta}{p}=\underset{\beta'}{p}$) and the estimates of the parameters in x_γ (2.15), one concludes that $b_{\alpha'}b_\beta b_{\beta'}=O'(1)$. This fact and (2.16) yield $b_{\alpha'}b_{\beta'}=O'(s^3)$, and, upon using (2.17), deduce

(2.18)
$$c_{\alpha'}c_{\beta'}^2=O'(1)/s^3.$$

Substituting the expansions (2.10) and the estimates above, into the Hamiltonian (1.5) yields

$$\frac{1}{2}\langle\lambda_{\alpha'},\lambda_{\alpha'}\rangle c_{\alpha'}^2+\frac{1}{2}\langle\lambda_{\beta'},\lambda_{\beta'}\rangle c_{\beta'}^2+\langle\lambda_{\alpha'},\lambda_{\beta'}\rangle c_{\alpha'}c_{\beta'}$$
$$+\frac{1}{s}\left[\langle\lambda_{\alpha'},\lambda_{\beta}\rangle c_{\alpha'}+\langle\lambda_{\beta'},\lambda_{\beta}\rangle c_{\beta'}\right]=\frac{1}{2s^2}\left(\frac{4}{9\langle\alpha,\alpha\rangle}-\langle\lambda_{\beta},\lambda_{\beta}\rangle\right)+\dots;$$

this fact and the estimates (2.18), (upon rescaling variables) yield

$$(sc_{\alpha'})=O'(1),\qquad (sc_{\beta'})=O'(1),$$

and so, using (2.17), deduce

$$b_{\alpha'}=O'(s),\qquad b_{\beta'}=O'(s^2),$$

finishing the proof of case (ii). The remaining two cases are done in a similar fashion. To actually check the parameters $s,\dots,$ are holomorphic parameters, one notes they are defined in terms of holomorphic functions (near $D_\alpha\cap D_\beta$) or their derivatives in t, completing the proof of Proposition 3.

Let $\Delta'\subset\Delta$; the boundary $\partial\Delta'$ of Δ' is defined as

$$\partial\Delta'\equiv\{\beta\in\Delta\backslash\Delta',\exists\alpha\in\Delta'\text{ such that }A_{\alpha\beta}\neq 0\},\tag{2.19}$$

and the exterior Δ'' of Δ' as

$$\Delta''\equiv\Delta\backslash(\Delta'\cup\partial\Delta'),\tag{2.20}$$

yielding a partition of Δ:

$$\Delta=\Delta'\cup\partial\Delta'\cup\Delta'',$$

such that

$$A_{\alpha\beta}=0\quad\text{for}\quad\alpha\in\Delta',\quad\beta\in\Delta'',\quad\text{and}\quad\partial\Delta''\subset\partial\Delta'.\tag{2.21}$$

The invariant structure of the Toda system (2.1) reflects this partitioning as follows:

Proposition 4. *I. Given an extended root system Δ and $\Delta'\subset\Delta$; all (polynomial) invariants of the (periodic) Toda system X_Δ have the form*

$$P(I',I'',I^\partial)+\sum_{\alpha\in\partial\Delta'}x_\alpha\mathscr{F}_\alpha(x,y)\tag{2.22}$$

where P and $\mathscr{F}_\alpha$ are polynomials in their arguments; I' (resp. I″) *stands for a basis of generators of the polynomial invariants*[12] *of the system*[13] *$X_{\Delta'}$ (resp. $X_{\Delta''}$) and I^∂ stands for the space of linear (connecting) invariants of dimension $|\partial\Delta'|-1$:*

$$I^\partial=\left\{\sum_{\alpha\in\Delta}v_\alpha y_\alpha\ \text{such that}\ \sum_{\alpha\in\Delta}v_\alpha\alpha\perp\hat{\beta},\ \forall\beta\in\Delta'\cup\Delta''\right\}\tag{2.23}$$
$$=\left\{\begin{array}{c}\sum_{\alpha\in\Delta}v_\alpha y_\alpha\ \text{such that}\\ \sum_{\alpha\in\Delta\backslash\beta}v_\alpha\alpha\in\operatorname{span}\{\lambda_\alpha,\alpha\in\partial\Delta'\backslash\beta\}\subset\Lambda_w(\Delta\backslash\beta)\\ \text{for any }\beta\in\partial\Delta'\end{array}\right\}$$

---

[12] they are not invariants of X_Δ

[13] depending on (x_α,y_α), $\alpha\in\Delta'$ and given by $A_{\Delta'}$

II. The polynomials I', I'' and I^{∂} are holomorphic near D_α, $\alpha\in\Delta'\cup\Delta''$, and $\bigcap_{\alpha\in\Delta'\cup\Delta''} D_\alpha$, in short:

$$\left.\begin{array}{l} I',I'',I^{\partial}|_{D_\alpha(t),\,\alpha\in\Delta'\cup\Delta''} \\ I',I'',I^{\partial}|_{D_{\Delta'\cup\Delta''}(t)} \end{array}\right\} = \textit{Taylor series in } t.$$

Proof of I. First observe that the submanifold

$$\{x_\alpha=0,\alpha\in\partial\Delta'\}$$

is an invariant subsystem of the Toda equations (2.1), i.e. it is preserved by the flow X_Δ. Also it is elementary to see that

$$X_\Delta|_{\{x_\alpha=0,\alpha\in\partial\Delta'\}} = X_{\Delta'}\oplus X_{\Delta''}\oplus\{\dot y_\alpha=\sum_{\beta\in\Delta'\cup\Delta''} A_{\alpha\beta}x_\beta,\alpha\in\partial\Delta'\}, \tag{2.24}$$

i.e. the flows $X_{\Delta'}$ and $X_{\Delta''}$ decouple and the variables y_α, $\alpha\in\partial\Delta'$ are completely specified by the linear differential equations in the third summand of (2.24); the latter are expressed in terms of the solution of the systems $X_{\Delta'}$ and $X_{\Delta''}$. It follows that the invariants of the system $X_\Delta\cap\{x_\alpha=0,\alpha\in\partial\Delta'\}$ are generated by those of $X_{\Delta'}$, $X_{\Delta''}$ and the invariants of the linear system.

An expression $\Sigma v_\alpha y_\alpha$ will be invariant for

$$\dot y_\alpha=\sum_{\beta\in\Delta'\cup\Delta''} A_{\alpha\beta x_\beta},\qquad \alpha\in\partial\Delta' \tag{2.25}$$

if and only if

$$\left(\sum_{\alpha\in\Delta} v_\alpha y_\alpha\right)^{\cdot} = \sum_{\substack{\alpha\in\Delta\\ \beta\in\Delta'\cup\Delta''}} v_\alpha A_{\alpha\beta}x_\beta = \sum_{\substack{\alpha\in\Delta\\ \beta\in\Delta'\cup\Delta''}} v_\alpha\langle\alpha,\hat\beta\rangle x_\beta$$
$$=\sum_{\beta\in\Delta'\cup\Delta''}\left\langle\sum_{\alpha\in\Delta} v_\alpha\alpha,\hat\beta\right\rangle x_\beta=0$$

for arbitrary x_β, $\beta\in\Delta'\cup\Delta''$. This holds if and only if

$$\sum_{\alpha\in\Delta} v_\alpha\alpha\perp\hat\beta,\qquad \text{for all } \beta\in\Delta'\cup\Delta'';$$

remember the trivial invariant $\Sigma p_\alpha y_\alpha=0$. Therefore the system (2.25) has $|\partial\Delta'|-1$ genuine linear invariants; they together enable us to eliminate all the y_α, $\alpha\in\partial\Delta'$, which shows there are no more invariants but the ones above. Therefore any invariant $\bar H$ of $X_\Delta\cap\{x_\alpha=0,\alpha\in\partial\Delta'\}$ must have the form

$$\bar H=P(I',I'',I^{\partial}),\qquad P \text{ polynomial}.$$

Moreover, since X_Δ preserves the manifold $\{x_\alpha=0,\alpha\in\partial\Delta'\}$, any invariant H_Δ of X_Δ must lead, upon restriction to $\{x_\alpha=0,\alpha\in\partial\Delta'\}$, to an invariant $\bar H$ of $X_\Delta\cap\{x_\alpha=0,\alpha\in\partial\Delta'\}$; hence

$$H_\Delta-P(I',I'',I^{\partial})=\sum_{\alpha\in\partial\Delta'} x_\alpha\mathscr{F}_\alpha,$$

proving part I, except for the second equality in (2.23). This is done by viewing $\sum_{\alpha\in\Delta} v_\alpha\alpha=\sum_{\alpha\in\Delta\backslash\beta} v'_\alpha\alpha$ (upon eliminating a root $\beta\in\partial\Delta'$) as a member of the weight lattice of the Lie algebra with basis $\Delta\backslash\beta$ and using the first equality.

Proof of Part II. (1) The following is straightforward:

$$I'|_{D_{\{\alpha\}}(t)} = \text{Taylor series in } t, \alpha \in \Delta'',$$

because on the one hand I' is a polynomial in x_β and y_β, $\beta \in \Delta'$, and on the other hand x_β and y_β, $\beta \in \Delta'$, do not blow up along $D_{\{\alpha\}}(t)$, for $\alpha \in \Delta''$; indeed the only x and y's blowing up along $D_{\{\alpha\}}(t)$ are x_α and y_β with $A_{\alpha\beta} \neq 0$. By symmetry the same statement holds for $I''|_{D_{\{\alpha\}}(t)}$, $\alpha \in \Delta'$.

(2) One verifies, using the Laurent solutions $D_{\{\alpha\}}(t)$, that

$$I'|_{D_{\{\alpha\}}(t)} = \text{Taylor series in } t, \alpha \in \Delta'$$

and by symmetry the same statement holds for $I''|_{D_{\{\alpha\}}(t)}$, $\alpha \in \Delta''$.

(3) To prove that

$$I^{\partial}|_{D_\beta(t),\, \beta \in \Delta' \cup \Delta''} = \text{Taylor series in } t,$$

observe

$$y_\alpha|_{D_\beta(t)} = -\frac{A_{\alpha\beta}}{t} + O(1), \qquad \alpha \in \Delta$$

and so $\Sigma v_\alpha y_\alpha \in I^{\partial}$ satisfies

$$\begin{aligned}\Sigma v_\alpha y_\alpha|_{D_\beta(t)} &= -\sum_\alpha v_\alpha A_{\alpha\beta} t^{-1} + O(1)\\ &= -\langle \Sigma v_\alpha \alpha, \hat{\beta}\rangle t^{-1} + O(1)\\ &= O(1), \qquad \text{using (2.23)}.\end{aligned}$$

Since $D_{\{\alpha\}}$ is a Zariski open subset of D_α, it suffices to check the statements above to guarantee finiteness and thus holomorphy of I', I'' and I^{∂} in a neighborhood of $D_\alpha, \alpha \in \Delta' \cup \Delta''$. By Hartog's lemma, I', I'' and I^{∂} are holomorphic on $\bigcap_{\alpha \in \Delta' \cup \Delta''} D_\alpha$ as well.

In view of the next proposition, we need the following two lemmas:

Lemma 1. (Tangency). *Consider the hypersurface*

$$D = \{(t_1, \ldots, t_l) \in \mathbb{C}^l \text{ such that } f(t) = 0\} \subset \mathbb{C}^l, \tag{2.26}$$

the top form on D

$$\omega_i = dt_1 \wedge \ldots \wedge \widehat{dt_i} \wedge \ldots \wedge dt_l|_D \tag{2.27}$$

and the vector field $\partial/\partial t_i$ in $\mathbb{C}^l$. Then the tangent locus $\mathscr{L}$ of $\partial/\partial t_i$ to D equals

$$\begin{aligned}\mathscr{L} &\equiv \left\{p \in D \text{ such that } \left.\frac{\partial}{\partial t_i}\right|_p \in T_p D\right\}\\ &= \{p \in D \text{ such that } \omega_i(p) = 0\}\\ &= \{p \in D \text{ such that } \varphi_{ij}(p) = 0\}\end{aligned} \tag{2.28}$$

with

$$\begin{aligned}\left\{\begin{matrix}\text{order of tangency to } D\\ \text{of } \partial/\partial t_i \text{ at } p \in D\end{matrix}\right\} &= \left\{\begin{matrix}\text{order of vanishing}\\ \text{of } \omega_i \text{ at } p \in D\end{matrix}\right\}\\ &= \left\{\begin{matrix}\text{order of vanishing}\\ \text{of } \varphi_{ij} \text{ at } p \in D\end{matrix}\right\}.\end{aligned}$$

In (2.28),

$$\varphi_{ij} \equiv \frac{\omega_i}{\omega_j} = (-1)^{i+j} \left.\frac{\partial f/\partial t_i}{\partial f/\partial t_j}\right|_D = (-1)^{i+j} \left.\frac{\partial f^n/\partial t_i}{\partial f^n/\partial t_j}\right|_D, \tag{2.29}$$

where $n \geqq 1$ *is an arbitrary integer.*

Proof. The obvious relation

$$0 = df|_D = \sum_i \frac{\partial f}{\partial t_i} dt_i|_D$$

enables one to express any $dt_i|_D$ in terms of the remaining $dt_j|_D$, yielding

$$\omega_i = (-1)^{i+j} \left.\frac{\partial f/\partial t_i}{\partial f/\partial t_j}\right|_D \omega_j\,;$$

this establishes (2.29). Then, using elementary calculus[14], we have

$$\begin{aligned}\{p, \partial/\partial t_i \in T_p D\} &= \{p \in D \text{ such that } \langle \text{grad } f(p), e_i\rangle = 0\}\\ &= \{p \in D \text{ such that } \frac{\partial f}{\partial t_i}(p) = 0\}\\ &= \{p \in D \text{ such that } \omega_i(p) = 0\}.\end{aligned}$$

Lemma 2 (Invariants). *The invariants H_1 and H_2 of the non-periodic Toda system with $\Delta = \{\alpha, \beta\}$*

$$\left\{\begin{array}{l} \dot{x}_\alpha = x_\alpha y_\alpha \quad \begin{pmatrix} y_\alpha \\ y_\beta \end{pmatrix}^{\cdot} = A_\Delta \begin{pmatrix} x_\alpha \\ x_\beta \end{pmatrix} \\ \dot{x}_\beta = x_\beta y_\beta \end{array}\right\}$$

are as follows: (set $y_\alpha^2 - 4x_\alpha \equiv \varphi_\alpha$)

$a_1 \oplus a_1$	$\overset{\bullet}{\alpha} \quad \overset{\bullet}{\beta}$	$A_\Delta = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$	$H_1 = \varphi_\alpha + \varphi_\beta$ $H_2 = \varphi_\alpha$
a_2	$\overset{\bullet}{\alpha}\!-\!\!-\!\overset{\bullet}{\beta}$	$A_\Delta = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$	$H_1 = y_\alpha^2 + y_\beta^2 + y_\alpha y_\beta - 3(x_\alpha + x_\beta)$ $H_2 = (2y_\alpha + y_\beta)(-y_\alpha + y_\beta)(y_\alpha + 2y_\beta)$ $+9x_\alpha(y_\alpha + 2y_\beta) - 9x_\beta(y_\alpha + y_\beta)$
c_2	$\overset{\bullet}{\alpha} \Rightarrow \overset{\bullet}{\beta}$	$A_\Delta = \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$	$H_1 = \varphi_\alpha + (y_\alpha + 2y_\beta)^2 - 8x_\beta$ $H_2 = \varphi_\alpha(y_\alpha + 2y_\beta)^2$ $+8x_\beta[y_\alpha(y_\alpha + 2y_\beta) + 2x_\beta)$
g_2	$\overset{\bullet}{\alpha} \Rrightarrow \overset{\bullet}{\beta}$	$A_\Delta = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$	$H_1 = (y_\alpha + 2y_\beta)^2 + 3\varphi_\beta - 4x_\alpha$ $H_2 = [((y_\alpha + 2y_\beta)(y_\alpha + y_\beta) + x_\beta)y_\beta$ $+x_\alpha(y_\alpha + 2y_\beta)]^2$ $-4x_\beta[(y_\alpha + 2y_\beta)(y_\alpha + y_\beta) + x_\beta]^2$ $+4x_\alpha x_\beta[(y_\alpha + 2y_\beta)^2 - x_\beta]$

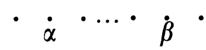
[14] $e_i = (0, \ldots, \underset{\substack{\uparrow \\ i}}{1}, 0, \ldots, 0)$

Proof. This is an immediate consequence of the table of invariants in Theorem 4, upon setting $x_0=0$ and eliminating y_0, using $p_0y_0+p_1y_1+p_2y_2=0$.

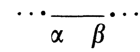
Proposition 5. *The Toda vector field*

$$X_\Delta : \dot{x}_\alpha = x_\alpha y_\alpha, \quad \dot{y}=A_\Delta x, \quad (\alpha\in\Delta \text{ and } \Sigma p_\alpha y_\alpha=0) \tag{2.1}$$

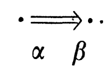
is transversal or tangent to D_α or D_β along the generic points of $D_\alpha\cap D_\beta\subset T^l$, according to the classification of Proposition 3. The tangency is governed by the basis $\Delta'=\{\alpha,\beta\}$; to wit, along $D_\alpha\cap D_\beta$

Case (i).


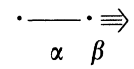
X_Δ is transversal to both D_α and D_β,

Case (ii).


X_Δ is simply tangent to both D_α and D_β,

Case (iii).


(a) (1) *X_Δ is doubly tangent to D_α,*
    (2) *X_Δ is transversal to D_α*
(b) *X_Δ is doubly tangent to D_β,*

Case (iv).


(a) *X_Δ is 4-fold tangent to D_α,*
(b) (1) *X_Δ is 4-fold tangent to D_β,*
    (2) *X_Δ is transversal to D_β.*

Proof of Proposition 5. We apply Lemma 1 to $D=D_\alpha$, $D\equiv D_\alpha=\{x_\alpha^{-1/2}=0\}\subset T^l=\mathbb{C}^l/\Lambda$, with coordinates $(t_1,\dots,t_l)\in\mathbb{C}^l$, such tat

$$\frac{\partial}{\partial t_1} \text{ corresponds to the Toda flow } X_\Delta, \text{ generated by } H_1$$

and

$$\frac{\partial}{\partial t_2} \text{ corresponds to the flow generated by } H_2 \text{ (of weighted degree } m_2+1).$$

By Lemma 1 and the Hamiltonian structure (1.14), the order of tangency of $\partial/\partial t_1 = X_\Delta$ to D_α at $p\in D_\alpha$ is given by the degree of vanishing at p of the following function defined on D_α:

$$\varphi_{12} = -\left.\frac{\partial x_\alpha/\partial t_1}{\partial x_\alpha/\partial t_2}\right|_{D_\alpha} = -\left.\frac{x_\alpha y_\alpha}{x_\alpha\left\langle \mu_\alpha, \dfrac{\partial H_2}{\partial y}\right\rangle}\right|_{D_\alpha} \tag{2.30}$$

Finding the degree of vanishing of $\varphi_{12}(p)$ along $D_\alpha\cap D_\beta$ is done in three steps:

1. substitute the solution (1.34) $D_{\{\alpha\}}(t)$ into φ_{12}, as given by (2.30)
2. take the limit when $t\to 0$, which yields $\varphi_{12}(p)$ of (2.29)
3. take the limit when $p\to p'\in D_\alpha\cap D_\beta$, yielding $\lim_{p\to p'}\varphi_{12}(p)$.

In order to estimate $\partial H_2/\partial y$ in (2.30), we apply Proposition 4 to the partition

$$\Delta = \Delta' \cup \partial\Delta' \cup \Delta'', \qquad \text{with } \Delta' = \{\alpha, \beta\}.$$

Moreover, since both expressions y_α and $\left\langle \mu_\alpha, \dfrac{\partial H_2}{\partial y} \right\rangle$ are logarithmic derivatives we have

$$y_\alpha|_{D_{\{\alpha\}}(t)} = \left.\frac{\partial x_\alpha/\partial t_1}{x_\alpha}\right|_{D_{\{\alpha\}}(t)} = -\frac{2}{t} + \ldots, \qquad \text{(confirming (1.28))}$$

and

$$\text{(2.31)} \quad \left.\left\langle \mu_\alpha, \frac{\partial H_2}{\partial y} \right\rangle\right|_{D_{\{\alpha\}}(t)} = \left.\frac{\partial x_\alpha/\partial t_2}{x_\alpha}\right|_{D_{\{\alpha\}}(t)} = -\frac{2}{t_2} + \ldots = \frac{\operatorname{Res}_t \left\langle \mu_\alpha \dfrac{\partial H_2}{\partial y} \right\rangle}{t} + \ldots.$$

Therefore

$$\text{(2.32)} \qquad \varphi_{12}^{-1}(p) = -\left.\frac{\partial x_\alpha/\partial t_2}{\partial x_\alpha/\partial t_1}\right|_{D_\alpha}, \qquad p \in D_\alpha$$

$$= -\left.\frac{x_\alpha \langle \mu_\alpha, \partial H_2/\partial y \rangle}{x_\alpha y_\alpha}\right|_{D_\alpha} \qquad \text{using (1.14), with}^{15}\ \mu_\alpha = \delta_\alpha^{-1} A_\alpha$$

$$= -\lim_{t \searrow 0} y_\alpha^{-1} \left.\left\langle \mu_\alpha, \frac{\partial H_2}{\partial y} \right\rangle\right|_{D_\alpha(t)} \qquad \text{using (2.31)},$$

$$= \frac{1}{2} \operatorname{Res}_t \left.\left\langle \mu_\alpha, \frac{\partial H_2}{\partial y} \right\rangle\right|_{D_\alpha(t)}$$

$$= \frac{1}{2} \operatorname{Res} \left.\left\langle \mu_\alpha, \frac{\partial}{\partial y} \left\{ P(I', I'', I^\partial) + \sum_{\gamma \in \partial\Delta'} x_\gamma \mathscr{F}_\gamma \right\} \right\rangle\right|_{D_\alpha(t)}$$

$$= \frac{1}{2} \operatorname{Res}_t \Bigg(\sum_j \partial P/\partial I_j'(I', I'', I^\partial) \langle \mu_\alpha, \partial I_j'/\partial y \rangle|_{D_\alpha(t)}$$

$$+ \sum_j \partial P/\partial I_j''(I', I'', I^\partial) \langle \mu_\alpha, \partial I_j''/\partial y \rangle|_{D_\alpha(t)}$$

$$+ \sum_j \partial P/\partial I_j^\partial(I', I'', I^\partial) \langle \mu_\alpha, \partial I_j^\partial/\partial y \rangle|_{D_\alpha(t)}$$

$$+ \sum_{\gamma \in \partial\Delta'} x_\gamma \langle \mu_\alpha, \partial \mathscr{F}_\gamma/\partial y \rangle|_{D_\alpha(t)} \Bigg)$$

$$= \frac{1}{2} \operatorname{Res}_t \Bigg(\sum \partial P/\partial I_j'(I', I'', I^\partial) \langle \mu_\alpha, \partial I_j'/\partial y \rangle|_{D_\alpha(t)}$$

$$+ \sum_{\gamma \in \partial\Delta'} x_\gamma \langle \mu_\alpha, \partial \mathscr{F}_\gamma/\partial y \rangle|_{D_\alpha(t)} \Bigg).$$

[15] $A_\alpha = (A_{1,\alpha}, \ldots, A_{l,\alpha})$

The last equality follows from the following facts:

$$\frac{\partial P}{\partial I}(I',I'',I^{\partial})_{D_{\{\alpha\}}(t)} \quad \text{where } I=I' \text{ or } I'' \text{ or } I^{\partial}$$
$$=\text{polynomial in } (I',I'',I^{\partial})|_{D_{\{\alpha\}}(t)}$$
$$=\text{Taylor series in } t \quad (\text{using Proposition 4, II});$$

$$\left\langle \mu_\alpha, \frac{\partial I''}{\partial y}\right\rangle=\delta_\alpha^{-1}\sum_\gamma A_{\gamma\alpha}\frac{\partial I''}{\partial y_\gamma}\equiv 0,$$

since I'' is a function of x_γ, y_γ, $\gamma\in\Delta''$ and since $A_{\gamma\alpha}=0$ for all $\gamma\in\Delta''$ (by (2.21)); and finally

$$\left\langle \mu_\alpha, \frac{\partial I^{\partial}}{\partial y}\right\rangle=\text{numerical constant, since } I^{\partial} \text{ is linear}.$$

Therefore according to (2.32), we need to estimate only two expressions.

Estimate 1.

$$\operatorname{Res}_t\left(\sum_{\gamma\in\partial\Delta'} x_\gamma\left\langle \mu_\alpha, \frac{\partial \mathscr{F}_\gamma}{\partial y}\right\rangle\right)\Bigg|_{D_\alpha(t)}=O'(1), \qquad \text{when } p\to p'\in D_\alpha\cap D_\beta{}^{16},$$

The weighted degree m_2+1 of H_2 is 3 for a_l, 4 for b_l, c_l, d_l, 5 for e_6, 6 for e_7, f_4, g_7 and 8 for e_8. Therefore using (2.22), the expressions $\mathscr{F}_\alpha$ have degrees m_2-1 and thus $\partial\mathscr{F}_\gamma/\partial y$ degrees m_2-2. Therefore, by Theorem 3:

$$\left\langle \mu_\alpha, \frac{\partial \mathscr{F}_\gamma}{\partial y}\right\rangle=t^{2-m_2}(R_0+R_1(b,c)t+R_2(b,c)t^2+\dots), \tag{2.33}$$

with $R_i(b,c)$ having weight i in b and c. Observe for $\gamma\in\partial\Delta'$,

$$x_\gamma=b_\gamma t+\dots \quad \text{or} \quad x_\gamma=b_\gamma+\dots \tag{2.34}$$

with

$$\begin{aligned} b_\gamma &\leqq O'(1) \quad \text{in cases (i), (ii) and (iii)}\\ &\leqq O'(s^2) \quad \text{in case (iv)}, \end{aligned} \tag{2.35}$$

according to Proposition 3. Therefore, since in cases (i), (ii) and (iii) $m_2+1=3$ or 4, we have, by (2.33), (2.34), (2.35) and $R_0=$constant, that

$$x_\gamma\left\langle \mu_\alpha, \frac{\partial \mathscr{F}_\gamma}{\partial y}\right\rangle\leqq\frac{O'(1)}{t},$$

which covers the cases a_l, b_l, c_l and d_l. In case (iv) where $m_2+1=6(2-m_2=-3)$, we have

$$\begin{aligned} \operatorname{Res} x_\gamma\left\langle \mu_\alpha, \frac{\partial \mathscr{F}_\gamma}{\partial y}\right\rangle &=b_\gamma\tilde{R}_1(b,c), \qquad \text{when } x_\gamma=b_\gamma t(1+c_\gamma t+\dots)\\ &=b_\gamma\tilde{R}_2, \qquad \text{when } x_\gamma=b_\gamma(1+c_\gamma t+d_\gamma t^2+\dots), \end{aligned}$$

[16] $O'(1)$ signifies growth in s

but according to case (iv) of Proposition 3

$$b_\gamma \leqq s^2 O'(1) \quad \text{and} \quad R_i(b,c) \leqq \frac{O'(1)}{s^i},$$

which implies

$$\operatorname{Res} x_\gamma \left\langle \mu_\alpha, \frac{\partial \mathscr{F}_\gamma}{\partial y} \right\rangle \leqq O'(1).$$

This covers the Lie algebras $\mathscr{g}_2^{(1)}$ and $\mathscr{d}_4^{(3)}$. Similar arguments hold for the remaining cases.

Estimate 2.

$$\operatorname{Res}_t \sum_{I_j'} \frac{\partial P}{\partial I_j'}(I', I'', I^\partial) \left\langle \mu_\alpha, \frac{\partial I_i'}{\partial y} \right\rangle \Bigg|_{D_\alpha(t)} \tag{2.36}$$

$$\begin{cases} = \dfrac{O'(1)}{s^{m_2'-1}} & \text{for } p \text{ near } p' \in D_\alpha \cap D_\beta \text{ in all cases} \\ & \text{of Proposition 3 not labeled (2)} \\ \leqq O'(1) & \text{for } p \text{ near } p' \in D_\alpha \cap D_\beta \\ & \text{in all cases labeled (2)}. \end{cases}$$

Since $\Delta' = \{\alpha, \beta\}$, the sum in the expression above ranges over the two invariants I_1' and I_2', listed in Lemma 2; they are extensions of the Weyl invariants going with Δ' and have weighted degrees $m_1'+2$ and $m_2'+1$. Moreover, by Proposition 4, II, the expressions I', I'', I^∂ have finite limits as $p \to p' \in D_\alpha \cap D_\beta$ and so does $\partial P/\partial I_i'$:

$$c_i(p') = \lim_{p \to p' \in D_\alpha \cap D_\beta} \frac{\partial P}{\partial I_i'}(I', I'', I^\partial).$$

In all four cases, one checks $c_2(p') \neq 0$, as $\partial P/\partial I_2'$ is either constant or a linear function of the y_i's; moreover, by adding a power of H_1 to H_2 (H_2 is not canonical), we can always assume $c_1(p') \neq 0$. Thus we have:

$$\begin{aligned} &\operatorname{Res}_t \sum_{I_j'} \frac{\partial P}{\partial I_j'}(I', I'', I^\partial) \left\langle \mu_\alpha, \frac{\partial I_i'}{\partial y} \right\rangle \Bigg|_{D_\alpha(t)} \\ &= \operatorname{Res}_t \sum_{I_j'} \frac{\partial P}{\partial I_j'}(I', I'', I^\partial) \delta_\alpha^{-1} \sum_{\gamma=\alpha,\beta} A_{\gamma\alpha} \frac{\partial I_i'}{\partial y_\gamma} \Bigg|_{D_\alpha(t)} \\ &\simeq \delta_\alpha^{-1} \sum_{i=1,2} c_i(p') \operatorname{Res}_t \left(2\frac{\partial I_i'}{\partial y_\alpha} + A_{\beta\alpha} \frac{\partial I_i'}{\partial y_\beta} \right) \Bigg|_{D_\alpha(t)} \\ &\simeq \delta_\alpha^{-1} (c_1(p') R^{(m_1'-1)}(b,c) + c_2(p') R^{(m_2'-1)}(b,c)), \end{aligned}$$

where (by Theorem 3) the $R^{(m_i'-1)}(b,c)$ are weighted expressions in the parameters b and c of weight $m_i'-1$; of course $m_1'-1=0$ in all cases and $m_2'-1=0,1,2,4$ in cases (i), (ii), (iii) and (iv) respectively. It therefore follows that at worst:

$$\begin{cases} i=1 & R^{(m_i-1)}(b,c)=R^{(0)}(b,c)=\text{rational constant} \neq 0 \\ i=2 & R^{(m_2-1)}(b,c) \end{cases}$$

$$\begin{cases} =\dfrac{O'(1)}{s^{m_2-1}}, & p\to p'\in D_\alpha\cap D_\beta \\ & \text{in all cases not denoted by (2)} \\ \leqq O'(1), & p\to p'\in D_\alpha\cap D_\beta \\ & \text{in all cases denoted by (2)}. \end{cases}$$

In order to prove the equality in (2.36), we need to make an computation. From its weight-homogeneity, $R^{(m_2-1)}(b,c)$ must contain the following ingredients (with numerical constants n_i), in parallel with the classification of Proposition 3:

Case.

(i) $R^{(0)}(b,c)=n_1$

(ii) $R^{(1)}(b,c)=n_2 c_\beta^{(1)}$

(iii) $R_a^{(2)}(b,c)=n_3 b_\alpha^{(2)}+n_4 c_\beta^{(1)2}+[\sum n_\gamma b_\gamma^{(2)}+c_\beta^{(1)}(\sum n'_\gamma c_\gamma^{(1)})]_a^{(2)}$

$R_b^{(2)}(b,c)=n_5 b_\alpha^{(2)}+n_6 c_\beta^{(1)2}+[\sum n''_\gamma b_\gamma^{(2)}+c_\beta^{(1)}(\sum n'''_\gamma c_\gamma^{(1)})]_b^{(2)}$

(iv) $R_a^{(4)}(b,c)=n_7 b_\alpha^{(2)2}+n_8 b_\alpha^{(2)} c_\beta^{(1)2}+n_9 c_\beta^{(1)4}+n_{10} b_\beta^{(3)} c_\beta^{(1)}+[n_{11} b_\gamma^{(3)} c_\beta^{(1)}]_a^{(4)}$

$R_b^{(4)}(b,c)=n_{12} b_\alpha^{(2)2}+n_{13} b_\alpha^{(2)} c_\beta^{(1)2}+n_{14} c_\beta^{(1)4}$

$+[n_{15} b_\gamma^{(2)} c_\beta^{(1)2}+n_{16} b_\gamma^{(2)} b_\alpha^{(2)}+n_{17} b_\gamma^{(2)2}]_b^{(4)}.$

It follows immediately from the estimates in Proposition 3 that

$$R_a^{(2)}(b,c) \quad\text{and}\quad R_b^{(4)}(b,c)\leqq O'(1)$$

in the cases denoted by (2). In order to have $R^{(m_2-1)}=O'(1)s^{-m_2+1}$ hold in all other case of Proposition 3, where $b_\alpha^{(2)}$ and $c_\beta^{(1)}$ blow up according to their weights, and where the other $b_\gamma^{(2)}$ and $c_\gamma^{(1)}$ blow up less forcefully than their weights, accidents among the constants n_i must not occur. Using the local description in Proposition 3, we must verify

$$n_1,n_2,\frac{n_3}{3}+n_4,\frac{n_5}{12}+n_6,n_7+n_8+n_9+n_{10}\left(8\pm\frac{16}{\sqrt{3}}\right) \tag{2.37}$$

$$\text{and}\quad \frac{n_{12}}{81}+\frac{n_{13}}{9}+n_{14}\quad \text{all} \neq 0.$$

Upon setting $I'_2=H_2$ of Lemma 2 and using the Laurent solution (1.34) $D_{\{\alpha\}}(t)$ to compute $R^{(m_2-1)}(b,c)$, we find

$$n_1=-4,\ n_2=36,\ n_3=0,\ n_4=-16,\ n_5=9\cdot 16,\ n_6=16,\ [\ ]_a^{(2)}=0 \quad\text{and}\quad [\ ]_b^{(2)}=0,$$

and so the first 4 inequalities of (2.37) certainly do hold. The fifth holds as well, as the n_i are always rational numbers (as follows from the recursion formula (2.4) for the coefficients of the solution $D_{\{\alpha\}}(t)$. The last inequality can be checked as well. This ends the verification of estimate 2.

To conclude, only *estimate 2* contributes to $\varphi_{12}(p)^{-1}$, which blows up as s^{-m_2+1} near $D_\alpha\cap D_\beta$, in all cases of Proposition 3 not denoted by (2). Therefore $\varphi_{12}(p)$

vanishes to orders s^{m_i-1} near $D_\alpha \cap D_\beta$, except in the cases (2), where it does not vanish. All these facts combined with Lemma 1 and the fact that s is a local parameter lead to Proposition 5.

§ 3. Linear equivalence and intersection of divisors (global theory)

This section deals with the proof of Theorems 1, 2 and 4; we begin with the *proof of Theorem1*. Proposition 1, stated and proved in § 2 is part of the statement of Theorem 1. Our attention now goes to the divisor D which completes the affine invariant manifold into an Abelian variety T^l. According to Theorem 3, each of the $l+1$ roots α of Δ maps into $2l+1$-parameter Laurent solutions, such that, among the x's, only x_α blows up as t^{-2}. Thus D consists of $l+1$ distinct components D_α, each one labeled by the unique x_α which blows up on it:

$$D=\sum_{\alpha\in\Delta} D_\alpha, \qquad \text{with } D_\alpha=\{x_\alpha^{-1}=0 \text{ on } T^l\}$$

and from the form of the Laurent solutions $D_{\{\alpha\}}(t)$ in (1.34), we have the divisor equality

$$(x_\alpha)=\sum_{\beta\in\Delta}(-A_{\alpha\beta})D_\beta, \qquad \alpha\in\Delta,$$

which confirms (1.17).

To prove (1.18) notice that the fundamental dominant weights $\lambda_1,\dots,\lambda_l$ relative to the semi-simple root system $\Delta_0=\{\alpha_1,\dots,\alpha_l\}=\Delta\backslash\{\alpha_0\}$ satisfy (1.8):

$$A_0\lambda=\alpha, \qquad \lambda=\begin{pmatrix}\lambda_1\\ \vdots\\ \lambda_l\end{pmatrix}, \qquad \alpha=\begin{pmatrix}\alpha_1\\ \vdots\\ \alpha_l\end{pmatrix}.$$

Inverting this system over $\mathbb{Z}$ yields

$$\gamma_i\lambda_i=\sum r_{ij}\alpha_j, \qquad \text{with } r_{ij}\in\mathbb{Z}, \tag{3.1}$$

where $\gamma_i=[\lambda_i, \Lambda_r(\Delta_0)]>0$ is the *smallest* positive integer such that $\gamma_i\lambda_i\in\Lambda_r(\Delta_0)$. The computation below is now performed in the dual Abelian variety Pic (T^l); there by (1.17)

$$A\begin{pmatrix}D_0\\ \vdots\\ D_l\end{pmatrix}\simeq\begin{pmatrix}(x_0)\\ \vdots\\ (x_l)\end{pmatrix}\simeq 0 \tag{3.2}$$

which is equivalent to, upon solving for the D_i in terms of D_0,

$$A_{\Delta_0}\begin{pmatrix}D_0\\ \vdots\\ D_l\end{pmatrix}\simeq\begin{pmatrix}0\\ \vdots\\ \boxed{\varepsilon D_0}\\ \vdots\\ 0\end{pmatrix}\to k^{\text{th}} \text{ place}, \tag{3.3}$$

with $\varepsilon=A_{0k}=\langle\alpha_0,\hat{\alpha}_k\rangle\in\mathbb{Z}, \neq 0$.

As a consequence of the formal identity (3.1), the equation (3.3) in $\operatorname{Pic}(T^l)$ has the solution

$$\gamma_i D_i \simeq \varepsilon r_{ik} D_0, \quad i=1,\dots,l. \tag{3.4}$$

Substituting this solution in (3.2) yields ($\gamma_0 = 1$)

$$\begin{aligned}
0 &\simeq \prod_0^l \gamma_i A \begin{pmatrix} D_0 \\ \vdots \\ D_l \end{pmatrix} \\
&\simeq A \prod \gamma_i \begin{pmatrix} \gamma_0^{-1} & & O \\ & \ddots & \\ O & & \gamma_l^{-1} \end{pmatrix} \begin{pmatrix} \gamma_0 D_0 \\ \vdots \\ \gamma_l D_l \end{pmatrix} \\
&\simeq A \prod \gamma_i \begin{pmatrix} \gamma_0^{-1} & & O \\ & \ddots & \\ O & & \gamma_l^{-1} \end{pmatrix} \begin{pmatrix} D_0 \\ \varepsilon r_{1k} D_0 \\ \vdots \\ \varepsilon r_{lk} D_0 \end{pmatrix}, \quad \text{(using (3.4))}.
\end{aligned}$$

Since each component of the last expression is a multiple of D_0 and equals 0 in $\operatorname{Pic}(T^l)$, we must have (after division by $\prod \gamma_i$)

$$A \begin{pmatrix} 1 \\ \varepsilon r_{1k}/\gamma_1 \\ \vdots \\ \varepsilon r_{lk}/\gamma_l \end{pmatrix} = 0,$$

and so by (1.9)

$$\left(1, \frac{\varepsilon r_{1k}}{\gamma_1}, \dots, \frac{\varepsilon r_{lk}}{\gamma_l}\right) = (\hat{p}_0, \hat{p}_1, \dots, \hat{p}_l), \quad \text{with } \hat{p}_0 = 1.$$

This relation combined with (3.4) yields (1.18), i.e.,

$$\gamma_i D_i \simeq \gamma_i \hat{p}_i D_0,$$

where we note that D_0 could have been replaced by another D_k.

In order to prove (1.16), we use relation (1.23) of Theorem 2 (which we prove later) and the following general statement about Abelian varieties T^l:

$$\dim L(D) = \frac{D^l}{l!} = g(D) - l + 1 = \prod_1^l \delta_i; \tag{3.5}$$

in this relation the divisor D on T^l defines a line bundle and an associated period matrix Ω_D and lattice Λ_D

$$T^l = \mathbb{C}^l/\Lambda_D, \quad \Omega_D = \left(\begin{array}{ccc|c} \delta_1 & & O & \\ & \ddots & & Z \\ O & & \delta_l & \end{array}\right) = \left\{\begin{array}{l}\text{period matrix} \\ \text{in appropriate} \\ \text{coordinates}\end{array}\right.$$

$$1 \leqq \delta_1 \leqq \dots \leqq \delta_l \in \mathbb{Z}, \quad \delta_i | \delta_{i+1}, \quad Z = Z^\top, \quad \operatorname{Im} Z > 0.$$

Then, setting $D=D_0$ and using (1.23), we find two different expressions for

$$g(D_0)-l+1=\prod_{\beta\in\Delta_0}\left(\frac{q_\beta}{\hat{p}_\beta}\right)=\prod_1^l \delta_i\,; \tag{3.6}$$

the reader is reminded of (1.2), namely that q_β are the integers appearing in the highest long root of $\mathscr{g}_0$. Referring to (1.2) and the Kac-Moody Lie algebra $L(\mathscr{g}, \nu)$, we now distinguish between the following three cases:

(i) $\nu=$identity; then $q_\beta=p_\beta$ and using $\delta_i|\delta_{i+1}$, we conclude

$$\delta_i=p_i/\hat{p}_i,$$

possibly after some relabeling.

(ii) ν^2 or $\nu^3=$identity and $L(\mathscr{g},\nu)\neq a_{2l}^{(2)}$; then $q_\beta=\hat{p}_\beta$ and thus

$$\prod_{\beta\in\Delta_0}\left(\frac{q_\beta}{\hat{p}_\beta}\right)=1$$
$$=\prod_1^l \delta_i \quad \text{using (3.6)},$$

implying that T^l is principally polarized, i.e.

$$\delta_1=\ldots=\delta_l=1\,.$$

(iii) $L(\mathscr{g},\nu)=a_{2l}^{(2)}$; if we use D_l in place of D_0, we find $\prod_1^l \delta_i=1$, leading to a principally polarized T^l as well.

To deduce (1.19), we apply the general relation

$$\dim L(\gamma D)=\frac{(\gamma D)^l}{l!}=\gamma^l\,\frac{D^l}{l!}=\gamma^l\dim L(D)\,,\qquad \gamma \text{ positive integer}$$

to $D\equiv\sum_0^l r_iD_i$ and $\gamma=\prod_0^l \gamma_i$ (defined earlier), yielding

$$\dim L\left(\sum_0^l r_iD_i\right)=\frac{1}{\gamma^l}\,\frac{\left(\gamma\sum_0^l r_iD_i\right)^l}{l!}$$
$$=\frac{1}{l!\gamma^l}\left(\sum_0^l\left(\frac{\gamma r_i}{\gamma_i}\right)\gamma_iD_i\right)^l,\quad \text{(trivially)},$$
$$=\frac{1}{l!\gamma^l}\left(\sum_0^l\left(\frac{r_i\gamma}{\gamma_i}\right)\gamma_1\hat{p}_iD_0\right)^l,\quad \text{using (1.18) and } \hat{p}_0=1\,,$$
$$=\left(\sum_0^l r_i\hat{p}_i\right)^l\frac{D_0^l}{l!}$$

$$=\left(\sum_0^l r_i\hat{p}_i\right)^l \dim L(D_0), \qquad \text{using (3.5)},$$

$$=\left(\sum_0^l r_i\hat{p}_i\right)^l \prod_1^l \delta_i\,; \qquad \text{using (3.5)}\,;$$

this leads to (1.19) using prior information, thus concluding the proof of Theorem 1.

Lemma 3. *On the Abelian variety T^l, we have $\bigcap_{\alpha\neq\alpha_0} D_\alpha=\{\text{one point}\}$.*

Proof. On each torus of the l-dimensional family of tori T^l, the intersection above is a finite $(\neq 0)$ number of points by (1.30) and the trajectory through these points is precisely given by the Laurent solution $D_{\Delta\setminus\alpha_0}(t)$:

(3.7) $x_i(t)=t^{-2}(\hat{n}_i+\ldots)$, $\qquad y_i(t)=t^{-1}(-2+\ldots)\quad i\neq 0$,

$$x_0(t)=t^{-2}\left(\prod_{i=1}^l \hat{n}_i^{-p_i}\right)(t^{2(p_0+\cdots+p_l)}+\ldots),\ y_0(t)=t^{-1}(2(p_1+\ldots+p_l)+\ldots).$$

These series depend on l free parameters $d_1,\ldots,d_l$, with d_i first occurring in the coefficient of t^{m_i+1} in the brackets; the m_i are the Weyl exponents of the Lie algebra generated by Δ_0. Moreover, as pointed out in Theorem 3, I, the coefficients of t^k (appearing in the brackets) are weight homogeneous polynomials of degree k in the d_i, upon specifying weight $(d_i)=m_i+1$.

To compute the trajectories through $\bigcap_{\alpha\neq\alpha_0} D_\alpha$ along a specific T^l, we confine the Laurent solutions $D_{\Delta\setminus\alpha_0}(t)$ to

$$T^l=\mathscr{A}\cup\Sigma D_\alpha\equiv\bigcap_0^l\{H_i=c_i\}\cup\Sigma D_\alpha,$$

and thus for $t\neq 0$ to the affine variety $\mathscr{A}$. This amounts to substituting the Laurent solution into each $H_i(x,y)=c_i$, $i=1,\ldots,l$; $H_0=\prod_0^l x_i^{p_i}=1$ has already been satisfied. Since the $H_i(x,y)$ are invariants of weighted degree m_i+1 (by Theorem 1), we have by Theorem 3, I

(3.8) $$H_i(x(t),y(t))=\frac{R_0(d)}{t^{m_i+1}}+\frac{R_1(d)}{t^{m_i}}+\ldots+R_{(m_i+1)}(d)t^0+\ldots\quad(i=1,\ldots,l)$$

$$=R_{(m_i+1)}(d_1,\ldots,d_i)\quad\text{(of weighted degree } m_i+1\text{)}$$

$$\overset{(*)}{=}k_id_i+p_{(m_i+1)}(d_1,\ldots,d_{i-1})\quad\text{for some } k_i\neq 0 \text{ and some polynomial } p_{(m_i+1)}\,;$$

the proof of $^{(*)}$ will be given later. Thus on generic $T^l=\mathscr{A}\cup D$,

$$1\leqq \#\Big(\bigcap_{\alpha\neq\alpha_0} D_\alpha\Big)\quad\Big(\text{since }\bigcap_{\alpha\neq\alpha_0} D_\alpha \text{ is non-empty}\Big) \tag{3.9}$$

$$= \#\{\text{trajectories } D_{\Delta\setminus\alpha_0}(t) \text{ confined to } T^l\}$$

$$= \#\bigcap_{i=0}^{l}\{H_i(x(t),y(t))=c_i\}$$

$$= \#\bigcap_{i=0}^{l}\{k_i d_i + p_{(m_i+1)}(d_1,\dots,d_{i-1})=c_i\},\qquad \text{using (3.8)}$$

$$\leqq 1,\qquad \text{since this relation is uniquely invertible}.$$

To prove $^{(*)}$, we proceed by induction, observing that by (3.9), the variety $\bigcap_1^l \{R_{(m_i+1)}(d)=c_i\}$ is non-empty for generic choice of $c_1,\dots,c_l$. Assume $R_{(m_i+1)}=k_i d_i+p_{(m_i+1)}(d_1,\dots,d_{i-1})=c_i$, with $k_i\neq 0$ for $i\leqq j$, where $j\geqq 0$; then $R_{(m_i+1)}(d)=c_i$, $k_i\neq 0$, $i\leqq j$ implies that $d_i=d_i(c_1,\dots,c_i)$, $i\leqq j$, and then $R_{(m_{(j+1)}+1)}(d)=c_{j+1}$, $k_{j+1}=0$ would yield a relation of the form

$$p_{m_{(j+1)}+1}(d_1(c_1),\dots,d_j(c_1,\dots,c_j))=c_{j+1}\quad (j\geqq 0),$$

which contradicts the fact that $c_1,\dots,c_l$ takes on arbitrary (generic) values.

Remark. One is tempted to use the same argument as before on $\bigcap_{\alpha\neq\beta} D_\alpha$. The generic solution $D_{\Delta\setminus\beta}(t)$ through this locus would lead to

$$x_i(t)=t^{-2}(\hat{n}_i+\dots),\qquad i\neq\beta$$

$$x_\beta(t)=t^{-2}\Big(\prod_{\alpha\neq\beta}\hat{n}_\alpha^{-p_\alpha}\Big)^{1/p_\beta}(t^{2(p_0+\dots+p_l)/p_\beta}+\dots).$$

Using Bézout's lemma, we would then have

$$\#\bigcap_{\alpha\neq\beta} D_\alpha\leqq p_\beta\frac{\prod(m_i(\Delta_0)+1)}{\prod(m_i(\Delta_\beta)+1)}=p_\beta\frac{|W_{\Delta_0}|}{|W_{\Delta_\beta}|},\qquad \text{by (1.13)},$$

with equality unless "accidents" occur. Since equality will be shown later, such accidents *do not* occur; however it seems hard to exclude them using a direct argument.

Proof of Theorem 2. Except for items (1.16) and (1.19), Theorem 1 has been established without using Theorem 2; thus we are free to use this part of Theorem 1 in deriving Theorem 2. For simplicity we give the proof of (1.20) for $\Delta'=\{\alpha_1,\alpha_2\}$, which amounts to checking Table 1, but first we show

Step 1. (1.20) implies (1.21), (1.22) and (1.23).
Indeed, Theorem 1 (not including (16) and (19)) implies

$$\mathscr{A}=\cap\{H_i=c_i\}=T^l\setminus\sum_{\alpha\in\Delta} D_\alpha,\qquad \begin{cases} T^l \text{ Abelian variety}\\ D_\alpha=\{x_\alpha^{-1}=0\}\cap T^l,\end{cases}$$

and (1.30) implies $\bigcap_{\beta\neq\alpha} D_\beta$ is a finite number n_α of points. We now evaluate the intersection below in two different ways ($\Delta_k \equiv \Delta \backslash \alpha_k$)

$$(3.10) \qquad D_0 \cdot D_1 \cdot D_2 \cdot \ldots \cdot \hat{D}_\beta \cdot \ldots \cdot D_l = n_\beta \frac{|W_{\Delta_\beta}|}{\det(A_{\Delta_\beta})}, \qquad \text{using (1.20)}$$

$$= \frac{\prod_1^l \hat{p}_i}{\hat{p}_\beta} D_0^l ;$$

the latter follows from relation (1.18), applied to

$$(3.11) \qquad \left(\prod_1^l \gamma_i\right) D_0 \cdot D_1 \cdot \ldots \cdot \hat{D}_\beta \cdot \ldots \cdot D_l = \gamma_\beta \cdot D_0 \cdot (\gamma_1 D_1) \cdot \ldots \cdot \hat{D}_\beta \cdot \ldots \cdot (\gamma_l D_l)$$

$$= \gamma_\beta \cdot D_0 \cdot (\gamma_1 \hat{p}_1 D_0) \cdot \ldots \cdot \hat{D}_\beta \cdot \ldots \cdot (\gamma_l \hat{p}_l D_0)$$

$$= \left(\prod_1^l \gamma_i\right) \frac{\left(\prod_1^l \hat{p}_i\right)}{\hat{p}_\beta} D_0^l .$$

The relations (3.10) imply

$$(3.12) \qquad n_\beta = \frac{\left(\prod_1^l \hat{p}_i\right) \det A_{\Delta_\beta}}{\hat{p}_\beta |W_{\Delta_\beta}|} D_0^l$$

and so

$$(3.13) \qquad \frac{n_\alpha}{n_\beta} = \left(\frac{p_\beta \hat{p}_\beta}{\det A_{\Delta_\beta}}\right) \left(\frac{p_\alpha \hat{p}_\alpha}{\det A_{\Delta_\alpha}}\right)^{-1} \left(\frac{p_\alpha |W_{\Delta_\beta}|}{p_\beta |W_{\Delta_\alpha}|}\right)$$

$$= \frac{p_\alpha |W_{\Delta_\beta}|}{p_\beta |W_{\Delta_\alpha}|}, \qquad \text{using (1.10)}.$$

The setting $k=0$ and applying Lemma 3 leads to $n_0 = 1$; the derivation of (1.21) is complete.

To prove (1.22), we compute on the one hand

(3.14)

$$D_0 \cdot D_1 \cdot \ldots \cdot \hat{D}_\beta \cdot \ldots \cdot D_l = n_\beta \frac{|W_{\Delta_\beta}|}{\det A_{\Delta_\beta}}, \qquad \text{using the first part of (3.10)}$$

$$= \left(\frac{p_\beta}{p_0} \frac{|W_{\Delta_0}|}{|W_{\Delta_\beta}|}\right) \frac{W_{\Delta_\beta}}{\det A_{\Delta_\beta}}, \qquad \text{using (3.13)}$$

$$= \left(\frac{p_\beta \hat{p}_\beta}{\det A_{\Delta_\beta}}\right) \frac{W_{\Delta_0}}{p_0 \hat{p}_\beta}$$

$$= \frac{|W_{\Delta_0}|}{\det A_{\Delta_0}} \frac{\hat{p}_0}{\hat{p}_\beta}, \qquad \text{using (1.10)},$$

and on the other hand

$$(3.15) \qquad D_0 \cdot D_1 \cdot \ldots \cdot \hat{D}_\beta \cdot \ldots \cdot D_l = \prod_{\alpha \neq \beta} \hat{p}_\alpha \cdot D_0^l \quad \text{by (3.11)},$$

$$= \prod_{\alpha \neq \beta} \hat{p}_\alpha \cdot l! \cdot (g(D_0) - l + 1) \quad \text{(by (3.5))},$$

Finally comparing (3.14) and (3.15), we have

$$g(D_0)-l+1=\frac{|W_{\Delta_0}|}{l!\det A_{\Delta_0}}\cdot\frac{1}{\prod\limits_{\alpha\in\Delta}\hat{p}_\alpha}=\prod_{\alpha\in\Delta}\left(\frac{q_\alpha}{\hat{p}_\alpha}\right),$$

$$\text{(using (1.13) and } \hat{p}_0=1)$$

concluding the proof of (1.23).

Step 2. (1.20) will now be established for two divisors D_α and D_β; this is the content of Table 1. The discussion follows the four cases (i), (ii), (iii) and (iv) of Propositions 3 and 5.

Case (i). According to Theorem 3, I, in particular (1.29), the function $g_{\alpha\beta}=(x_\alpha x_\beta)^{-1/2}$ vanishes simply along D_α and D_β, it is finite and nonzero in the affine, since $H_0=\prod\limits_{\alpha\in\Delta}x_\alpha^{p_\alpha}=1$. Therefore in the neighborhood $U_{\alpha\beta}$ of the $m-2$-dimensional variety $D_{\alpha\beta}$, (see (1.24)), the function $g_{\alpha\beta}$ is holomorphic and vanishes along $D_{\{\alpha\}}\cup D_{\{\beta\}}$ only, i.e., $g_{\alpha\beta}|_{D_{\{\alpha\}}(t)}$ is a Taylor series in t, with holomorphic coefficients in the free parameters (s,b'_γ,c'_γ) of the $D_{\{\alpha\}}(t)$ series; their behavior can be deduced from Proposition 3. Upon confining $D_{\{\alpha\}}(t)$ to a fixed torus T^l, some of the (b'_γ,c'_γ) account for the values of the constants of the motion and others are running parameters describing D_α. Since by Proposition 5, (i) the Toda flow X_Δ is transversal to $D_{\{\alpha\}}$ (and $D_{\{\beta\}}$) near $D_{\alpha\beta}$, $D_{\{\alpha\}}$ and $D_{\{\beta\}}\subset T^l$ may behave near $D_{\alpha\beta}$ in many different ways, i.e., several sheets of $D_{\{\alpha\}}$ and $D_{\{\beta\}}$ may meet at the intersection $D_{\alpha\beta}$, and they may or may not be tangent to one another. Now using the estimates in the proof of Proposition 3, (i), compute

$$\begin{aligned}g_{\alpha\beta}|_{D_{\{\alpha\}}(t)}&=b_\beta^{-1/2}t(1-c_\beta t/2+\dots)\\&=2st+t^2+tO'_2\\&=(2s+t)t\cdot\text{unit}\quad\text{near }D_{\alpha\beta},\end{aligned}$$

where O'_2 stands for a holomorphic series, quadratic or more in t and s, whose coefficients depend on the (b'_γ,c'_γ). Since near $D_\alpha\cap D_\beta$ we have $g_{\alpha\beta}(t,s)=0$ precisely on $D_\alpha\cup D_\beta$, exactly one sheet of D_α must cross one sheet of D_β transversely at $D_\alpha\cap D_\beta$, leading to the picture advertised in case (i) of Table 1.

Case (ii). It proceeds along similar lines, with the formula (1.29) dictating the new choice:

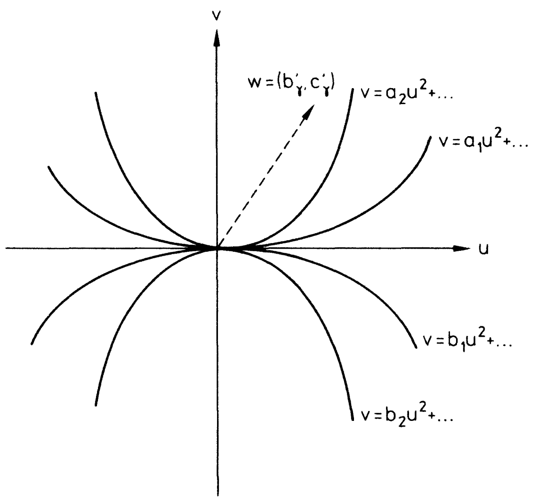
$$g_{\alpha\beta}=(x_\alpha x_\beta)^{-1},$$

using the estimates in the proof of Proposition 3, (ii), we again compute $g_{\alpha\beta}$ along the D_α trajectories near $D_\alpha\cap D_\beta$, making use of the local parametrization $(t,s,b'_\gamma,c'_\gamma)$, to wit:

$$\begin{aligned}(3.16)\qquad g_{\alpha\beta}|_{D_{\{\alpha\}}(t)}&=b_\beta^{-1}t\left(1-c_\beta t+(c_\beta^2-b_\alpha+b_{\beta'})\frac{t^2}{2}+f_\beta t^3+\dots\right)\\&=-\left(\frac{3s}{2}\right)^3t+\left(\frac{3}{2}\right)^3s^2t^2-\frac{3}{2}st^2+\frac{1}{4}t^4+tO'_4\end{aligned}$$

near $D_{\alpha\beta}$, using Proposition 3.

In case (ii) a new complication arises: by Proposition 5, (ii), the Toda vector field X_Δ is simply tangent to both $D_{\{\alpha\}}$ and $D_{\{\beta\}} \subset T^l$ along $D_{\alpha\beta}$; thus all sheets of $D_{\{\alpha\}}$ (say m) and $D_{\{\beta\}}$ (say n) must be simply tangent to the u-axis (direction X_Δ); hence, $D_{\{\alpha\}}$ and $D_{\{\beta\}}$ are at least simply tangent to each other at D_α. Let v be a transversal direction to u, making the uv-plane into a transversal slice of $D_{\{\alpha\}}$ and $D_{\{\beta\}}$ near $D_{\alpha\beta}$; let w account for the (b'_γ, c'_γ)-axes (see Fig.).



Since the function $g_{\alpha\beta}$ vanishes precisely on $D_{\{\alpha\}} \cup D_{\{\beta\}}$, the Weierstrass preparation theorem implies

$$g_{\alpha\beta} = \prod_{i=1}^{m} (v - a_i u^2) \cdot \prod_{j=1}^{m} (v - b_j u^2) \cdot \text{unit} \quad a_i, b_j \neq 0 .$$

The Toda flow moves an arbitrary point (u_0, v_0) on $D_{\{\alpha\}}$ (assume it is situated on the branch $v_0 = a_1 u_0^2 + \dots$) to $(u, v) = (u_0 + t, a_1 u_0^2 + \dots)$. Therefore $g_{\alpha\beta}$ evaluated along the trajectory near that branch takes on the form

$$\text{(3.17)} \quad g_{\alpha\beta}|_{D_{\{\alpha\}}(t)} = [t(t+2u_0) + O'_3] \prod_{i=2}^{m} \left[t(t+2u_0) + \left(1 - \frac{a_1}{a_i}\right) u_0^2 + O'_3 \right]$$

$$\prod_{j=1}^{n} \left[t(t+2u_0) + \left(1 - \frac{a_1}{b_j}\right) u_0^2 + O'_3 \right] \cdot \text{unit}$$

$$= \left(2 \prod_{2}^{m} \left(1 - \frac{a_1}{a_i}\right) \prod_{1}^{n} \left(1 - \frac{a_1}{b_j}\right) u_0^{2m+2n-1} t + \dots + t^{2m+2n} \right) \cdot \text{unit}$$

(Weierstrass polynomial in t and u_0).

Comparing the two expressions (3.16) and (3.17), one finds $m + n = 2$ and thus $m = n = 1$, since $m \geqq 1$ and $n \geqq 1$; one also finds that the coefficient of $u_0^{2m+2n-1} t$ in (3.17) does not vanish, and that $u_0 = s$. It also implies that the D_α and D_β sheets are simply tangent to each other near $D_\alpha \cap D_\beta$.

Case (iii). Proceeds as in the above cases by considering the function

$$g_{\alpha\beta} \equiv x_\alpha^{-3/2} x_\beta^{-2},$$

picked according to the recipe (1.29). Remember that according to Proposition 5, the flow X_Δ is transversal to at least one branch of D_α (case (iii, a, 2)). It is thus advantageous to use this branch and hence the corresponding estimates as given in Proposition 3:

(3.18)

$$g_{\alpha\beta}|_{D_\alpha(t)} = \frac{t}{b_\beta^2} + \ldots + ct^7 + \ldots \quad \text{for some function } c \text{ of the free parameters}$$

$$= k_1 s^2 t + k_2 s t^4 + k_3 t^7 + t O_7', \quad \text{for some constants } k_i, \text{ with } k_2^2 \neq 4 k_1 k_3 \text{ using the estimates of case (iii, a, 2)}.$$

Besides being transversal to at least k D_α-branches, the flow X_Δ is doubly tangent to at least l D_α-branches and m D_β-branches $(m, n \geqq 1)$. Since $g_{\alpha\beta} = 0$ along $D_\alpha \cup D_\beta$ only in the neighborhood of $D_{\alpha\beta}$, we must have

$$g_{\alpha\beta} = \prod_1^k (t - c_i s^{n_i}) \prod_1^l (s - a_i t^3 + \ldots) \prod_1^m (s - b_j t^3 + \ldots) \cdot \text{unit},$$

which, compared to (3.18), forces us to have $k + 3l + 3m = 7$ and thus $k = l = m = 1$. Since $k_2^2 \neq 4 k_1 k_3$, we have $a_i \neq b_j$, implying the two sheets (doubly tangent to the flow) are only doubly tangent to each other, confirming the statement for case (iii). *Case (iv)* is in the same style as before, only requiring more computations. From Proposition 5, case (iv), the divisor D_α has at least one branch which is 4-fold tangent to the flow X_Δ along $D_{\alpha\beta}$ (case (iv, a)) and D_β has two branches, one which is 4-fold tangent to X_Δ (case (iv, b, 1)) and a transversal one (case (iv, b, 2)). The function used here is $g_{\alpha\beta} = x_\alpha^{-3} x_\beta^{-5}$ (see (1.29)) which vanishes simply along D_α and D_β. Using the transversal branch and hence estimates of case (iv, b, 2), we find

$$g_{\alpha\beta}|_{D_\beta(t)} = \frac{t}{b_\alpha^3} + \ldots + ct^{16} + \ldots$$

$$= k_1 s^3 t + k_2 s^2 t^6 + k_3 s t^{11} + k_4 t^{16} + t O_{16}'$$

leading to the results advertised in Table 1. This ends the proof of Theorem 2.

Proof of Theorem 4. First observe that for the case of an Abelian surface T^2, the divisor D satisfies

$$\dim L(D) = \frac{D \cdot D}{2} = g(D) - 1 = \delta_1 \delta_2, \tag{3.19}$$

where

$$T^2 = \mathbb{C}^2/\Lambda, \quad \text{lattice } \Lambda \text{ generated by } \Omega = \left(\begin{array}{cc|c} \delta_1 & 0 & \\ & & Z \\ 0 & \delta_2 & \end{array} \right),$$

and where $g(D)$ is the genus of the divisor D.

Remembering from Theorem 1 the equivalence in $\operatorname{Pic}(T^2)$

$$\gamma_i D_i \simeq \gamma_i \hat{p}_i D_0, \tag{3.20}$$

we compute the intersection numbers

$$(\gamma_i D_i)\cdot(\gamma_j D_j)=(\gamma_i\hat{p}_i D_0)\cdot(\gamma_j\hat{p}_j D_0)=\gamma_i\gamma_j\hat{p}_i\hat{p}_j(D_0\cdot D_0),$$

which combined with (3.19) yields

$$\frac{D_i\cdot D_i}{2}=\hat{p}_i\hat{p}_j\frac{D_0\cdot D_0}{2}=\hat{p}_i\hat{p}_j(g(D_0)-1) \tag{3.21}$$

and, in particular

$$\begin{aligned}\frac{D_i\cdot D_i}{2}&=\hat{p}_i^2(g(D_0)-1)\\&=g(D_i)-1\quad\text{(using again (3.19))}.\end{aligned} \tag{3.22}$$

Therefore we have

$$\begin{aligned}g(D_0)-1&=\frac{D_i\cdot D_i}{2\hat{p}_i\hat{p}_j}, &&\text{using (3.21)},\\&=\frac{g(D_i)-1}{\hat{p}_i^2}, &&\text{using (3.22)},\\&=\delta_1\delta_2, &&\text{using (3.19)},\\&=\frac{q_1q_2}{\hat{p}_1\hat{p}_2}, &&\text{using (1.23)},\end{aligned}$$

leading to the following formula

$$g(D_i)-1=\hat{p}_i^2\,\frac{q_1q_2}{\hat{p}_1\hat{p}_2}\qquad i=0,1,2$$

$$D_i\cdot D_j=2\hat{p}_i\hat{p}_j\cdot\frac{q_1q_2}{\hat{p}_1\hat{p}_2}\qquad i,j=0,1,2$$

with the understanding that always $\hat{p}_0=1$. This yields Table 4.

Table 4

Lie Algebra	$(\hat{p}_1,\hat{p}_2)$	(q_1,q_2)	(δ_1,δ_2)	$g_{D_0}-1$	$g_{D_1}-1$	$g_{D_2}-1$	$D_0\cdot D_1$	$D_1\cdot D_2$	$D_0\cdot D_2$
$a_2^{(1)}$	(1, 1)	(1, 1)	(1, 1)	1	1	1	2	2	2
$c_2^{(1)}$	(1, 1)	(2, 1)	(1, 2)	2	2	2	4	4	4
$g_2^{(1)}$	(2, 1)	(2, 3)	(1, 3)	3	12	3	12	12	6
$a_4^{(2)}$	$(1,\frac{1}{2})$	(1, 2)	(2, 2)	4	4	1	8	4	4
$d_3^{(2)}$	(2, 1)	(1, 2)	(1, 1)	1	4	1	4	4	2
$d_4^{(3)}$	(2, 3)	(3, 2)	(1, 1)	1	4	9	4	12	6

By Theorem 2, the local behavior of $D_j\cap D_k$ is regulated by the Lie algebra generated by $\{\alpha_j,\alpha_k\}=\Delta_i\equiv\Delta\backslash\alpha_i$. If $D_j\cap D_k=$ sum of points, then, according to (1.20), we have

$$m_{jk}\equiv\operatorname{mult}_{\text{point}}(D_j\cdot D_k)=\frac{|W_{\Delta_i}|}{\det A_{\Delta_i}},$$

and

$$\#(D_j \cap D_k) = \frac{D_j \cdot D_k}{m_{jk}},$$

which together with Table 4 below yields Theorem 4.

Table 5. Intersection data for Abelian surfaces

Lie Algebra	Δ_2	Δ_0	Δ_1	$\#D_0 \cap D_1$	$\#D_1 \cap D_2$	$\#D_2 \cap D_0$
$a_2^{(1)}$	$m=$ $\underset{\alpha_0\qquad\alpha_1}{\cdot\overset{2}{\text{------}}\cdot}$	$\underset{\alpha_1\qquad\alpha_2}{\cdot\overset{2}{\text{------}}\cdot}$	$\underset{\alpha_0\qquad\alpha_2}{\cdot\overset{2}{\text{------}}\cdot}$	1	1	1
$c_2^{(1)}$	$m=$ $\underset{\alpha_0\qquad\alpha_1}{\cdot\overset{4}{\Longrightarrow}\cdot}$	$\underset{\alpha_1\qquad\alpha_2}{\cdot\overset{4}{\Longleftarrow}\cdot}$	$\underset{\alpha_0\qquad\alpha_2}{\cdot\overset{1}{\qquad}\cdot}$	1	1	4
$g_2^{(1)}$	$m=$ $\underset{\alpha_0\qquad\alpha_1}{\cdot\overset{2}{\text{------}}\cdot}$	$\underset{\alpha_1\qquad\alpha_2}{\cdot\overset{12}{\Rrightarrow}\cdot}$	$\underset{\alpha_0\qquad\alpha_2}{\cdot\overset{1}{\qquad}\cdot}$	6	1	6
$a_4^{(2)}$	$m=$ $\underset{\alpha_0\alpha_1}{\cdot\overset{4}{\Longrightarrow}\cdot}$	$\underset{\alpha_1}{\cdot\overset{4}{\Longrightarrow}\cdot}$	$\underset{\alpha_0\qquad\alpha_2}{\cdot\overset{1}{\qquad}\cdot}$	2	1	4
$d_3^{(2)}$	$m=$ $\underset{\alpha_0\qquad\alpha_1}{\cdot\overset{4}{\Longleftarrow}\cdot}$	$\underset{\alpha_1\qquad\alpha_2}{\cdot\overset{4}{\Longrightarrow}\cdot}$	$\underset{\alpha_0\qquad\alpha_2}{\cdot\overset{1}{\qquad}\cdot}$	1	1	2
$d_4^{(3)}$	$m=$ $\underset{\alpha_0\qquad\alpha_1}{\cdot\overset{2}{\text{------}}\cdot}$	$\underset{\alpha_1\qquad\alpha_2}{\cdot\overset{12}{\Lleftarrow}\cdot}$	$\underset{\alpha_0\qquad\alpha_2}{\cdot\overset{1}{\qquad}\cdot}$	2	1	6

§ 4. The Lie algebras $a_l^{(1)}$ and $c_l^{(1)}$ – specific results

By considering the specific structure of the Kac-Moody Lie algebras, one can obtain more detailed descriptions of the divisors and their intersection. Although the results below can be adapted to all the Kac-Moody Lie algebras, we limit ourselves here to $a_l^{(1)}$ and $c_l^{(1)}$.

1. The Toda lattice for $a_l^{(1)}$

Using the classical representation of $a_l^{(1)}$, the Lax pair (1.4) for the $a_l^{(1)}$ Toda lattice takes on the form $\dot{L} = [L, P]$, where $L = L_h$ is the matrix (after a slight change of y-coordinates)

$$L = L_h = \begin{pmatrix} y_1 & -x_1 & 0 & 0 & h^{-1} \\ 1 & y_2 & -x_2 & & \\ 0 & 1 & y_3 & \vdots & \vdots \\ \vdots & & & & \\ 0 & & & y_l & -x_l \\ -x_0 h & & & 1 & y_{l+1} \end{pmatrix}, \tag{4.1}$$

satisfying

$$\sum_1^{l+1} y_i = 0, \qquad H_0 = \prod_0^l x_i = c_0 = 1.$$

Then the (time-invariant) isospectral curve

$$\mathscr{C} = \left\{ \begin{matrix} (z,h) \text{ such that } \det(zI - L_h) = \pm(h + h^{-1}) + P_{l+1}(z) = 0 \\ \text{with } P_{l+1}(z) = z^{l+1} + H_1(x,y) z^{l-1} + \ldots + H_l(x,y) \end{matrix} \right\}$$

has genus $g = l$ and it is hyperelliptic, with hyperelliptic involution

$$(z,h) \rightsquigarrow (z, h^{-1})$$

and irrationality

$$y = \sqrt{P_{l+1}^2 - 4}.$$

The divisors of poles and zeroes of h and z are given by

$$(h) = (l+1)Q - (l+1)P, \qquad (z) \geqq -P - Q,$$

where P and Q are the two points covering ∞. This shows that $\mathscr{C}$ is a special hyperelliptic curve with so-called division points.

It is well-known that $\operatorname{Jac}(\mathscr{C})$ can be defined as

$$\operatorname{Jac}(\mathscr{C}) = (S^g \mathscr{C} / \equiv)$$

where $\equiv$ stands for the customary Abel identifications of divisors. Let W_k be the image of $S^k \mathscr{C}$ via the projection

$$S^g \mathscr{C} \to \operatorname{Jac}(\mathscr{C}).$$

Theorem 5. *For the $a_l^{(1)}$-Toda lattice, the affine part $\mathscr{A}$ completes into the Jacobian* $\operatorname{Jac}(\mathscr{C})$ *of the spectral curve upon adjoining $l+1$ translates of the theta-divisor Θ:*

$$\mathscr{A} = \operatorname{Jac}(\mathscr{C}) \backslash \sum_0^l (\Theta + k(Q-P)),$$

i.e. the divisors D_{α_k} of Theorem 1 are given by

$$D_{\alpha_k} = \Theta + k(Q-P) \quad k = 0, \ldots, l.$$

Moreover for $\Delta' = \{\alpha_{i_1}, \alpha_{i_2}, \ldots, \alpha_{i_k}\} \subset \Delta$

$$\textit{mult.}\,(D_{\alpha_{i_1}}) \cdot \ldots \cdot D_{\alpha_{i_k}}) = \prod_{\substack{\Delta_r'' \subset \Delta' \\ \Delta_r'' \text{ connected}}} (|\Delta_r''|!)$$

$$\textit{reduced}\,(D_{\alpha_{i_1}} \cdot \ldots \cdot D_{\alpha_{i^k}}) \sim \frac{k!}{\prod_{\substack{\Delta_r'' \subset \Delta' \\ \Delta_r'' \text{ connected}}} |\Delta_r''|!} W_{g-k} \quad \textit{(homologous)} \tag{4.2}$$

For two divisors, we have the stronger result:

(i) *two contiguous divisors satisfy:*

$$D_{\alpha_i} \cap D_{\alpha_{i+1}} = 2\, W_{g-2} \quad \textit{(modulo translation)}$$

(ii) *two non-contiguous divisors satisfy* ($k>1$):

$$D_{\alpha_i} \cap D_{\alpha_{i+k}} = \textit{irreducible variety} \sim 2W_{g-2} \quad \textit{(homologous)}.$$

Proof. At first we define the following isomorphism:

$$i:\{(x,y)\in\mathscr{A}\}\to\mathscr{A}'=\left\{\begin{matrix}\text{divisor } \mathscr{D} \text{ on } \mathscr{C} \text{ such that} \\ \mathscr{L}(\mathscr{D}+kP-(k+1)Q)=\phi,\ \forall k\in\mathbb{Z}\end{matrix}\right\}\subset \operatorname{Jac}(\mathscr{C}).$$

At first, a point $(x,y)\in\mathscr{A}$ maps into a matrix L_h and the associated eigenvector f:

$$L_h f = zf, \quad f=(f_1,\dots,f_{l+1})^{\top}, \quad f_i=\frac{\Delta_{1i}}{\Delta_{11}},$$

where $\Delta_{ij}=\Delta_{ij}(z,h)$ stands for the minor of $L_h - zI$ corresponding to the $(i,j)^{\text{th}}$ entry. Then there exists a minimal positive affine divisor $\mathscr{D}$ of order $g=l$ on the curve $\mathscr{C}$ such that

$$(f_k)\geqq -\mathscr{D}-(k-1)P+(k-1)Q, \quad k=1,\dots,l+1$$

with equality at the points P and Q covering $z=\infty$. One shows order $(\mathscr{D})=l$, $\mathscr{D}$ is non-special and

$$\dim \mathscr{L}(\mathscr{D}+kP-(k+1)Q)=0, \quad \text{for all } k\in\mathbb{Z}. \tag{4.3}$$

Conversely a divisor with these properties maps into a unique matrix L_h of the form (4.1). For details, see van Moerbeke and Mumford [23] and Adler and van Moerbeke [3]. In view of the fact that $(h)=(l+1)Q-(l+1)P$, it suffices to require (4.3) for $k=1,\dots,l+1$. Therefore

$$\operatorname{Jac}(\mathscr{C})\backslash\mathscr{A}'=\sum_{k=0}^{l} D_k$$

where

$$D_k\equiv\{\mathscr{D}\in\operatorname{Jac}(\mathscr{C}) \text{ such that } \dim L(\mathscr{D}+kP-(k+1)Q)>0\}.$$

Then D_0 ist the theta-divisor Θ, i.e. the image of $S^{g-1}\mathscr{C}$ in $\operatorname{Jac}(\mathscr{C})$ by means of the Abel map $S^g\mathscr{C}\to\operatorname{Jac}(\mathscr{C})$ (modulo some translation), i.e.,

$$\begin{aligned} D_0 &= \{\mathscr{D}\in\operatorname{Jac}(\mathscr{C}) \text{ such that } (f)\geqq -\mathscr{D}+Q \text{ for some } f\neq 0)\} \\ &= Q+W_{g-1} \\ &= \Theta \end{aligned}$$

and

$$\begin{aligned} D_k &= \{\mathscr{D}\in\operatorname{Jac}(\mathscr{C}) \text{ such that } (f)\geqq -\mathscr{D}+(k+1)Q-kP \text{ for some } f\neq 0\} \\ &= (k+1)Q-kP+W_{g-1} \\ &= \Theta+k(Q-P). \end{aligned}$$

To show $D_k=D_{\alpha_k}$, we prove equalities (i) and (ii) for D_k in place of D_{α_k}: For a general compact Riemann surface $\mathscr{C}$, let $\theta(z)$ be the Riemann theta function, and $\omega=(\omega_1,\dots,\omega_g)$ a basis of holomorphic differentials; then for all $a,b,c,d\in\mathscr{C}$ and

$\mathbf{z} \in \mathscr{C}^g$:

$$c_1 \theta\left(\mathbf{z} + \int_{a+b}^{c+d} \omega\right) \cdot \theta(z) = c_2 \theta\left(\mathbf{z} + \int_b^c \omega\right) \cdot \theta\left(\mathbf{z} + \int_a^d \omega\right)$$
$$+ c_3 \theta\left(\mathbf{z} + \int_a^c \omega\right) \theta\left(\mathbf{z} + \int_b^d \omega\right)$$

(Fay's trisecant identity).

This identity has a number of remarkable geometric consequences. One of them is that for $u, v \in W_1 \simeq \mathscr{C}$ and k the canonical divisor on $\mathscr{C}$, we have

$$W_{g-1} \cap (W_{g-1} + u - v) = (W_{g-2} + u) \cup (-W_{g-2} + k - v), \tag{4.4}$$

which we now apply to the hyperelliptic curve $\mathscr{C}$ of genus $g = l$ with u, v and k picked as follows (remember $y = (P_{l+1}^2 - 4)^{1/2}$):

$$k = (dz/y) = (l-1)P + (l-1)Q, \tag{4.5}$$
$$u = P, \quad v = Q.$$

Then

$$W_{g-1} = D_{r+1}, \quad W_{g-1} + u - v = D_{r+1} + (P - Q) = D_r$$

and the two sets on the right of (4.4) turn out to be equal:

$$W_{g-2} + P = -W_{g-2} + (g-1)P + (g-2)Q;$$

indeed, since for any choice of j points $p_i = (z_i, h_i) \in \mathscr{C}$

$$\left(\prod_1^j (z - z_i)\right) = \sum_1^j (p_i + p_i^\tau) - j(P + Q)$$

we have

$$\sum_1^j (p_i + p_i^\tau) = j(P + Q) \quad \text{in } \operatorname{Jac}(\mathscr{C}), \qquad \text{for all } p_i \in \mathscr{C}.$$

Therefore setting the origin at P and $j = g - 2$, we have

$$\sum_1^{g-2} \int_P^{p_i} \omega + \sum_1^{g-2} \int_P^{p_i^\tau} \omega = (g-2) \int_P^Q \omega$$

and thus

$$\sum_1^{g-2} \int_P^{p_i} \omega = -\sum_1^{g-2} \int_P^{p_i^\tau} \omega + (g-2) \int_P^Q \omega$$

with

$$\sum_1^{g-2} \int_P^{p_i} \omega \in W_{g-2} + P$$

and

$$-\sum_1^{g-2} \int_P^{p_i^\tau} \omega + (g-2) \int_P^Q \omega \in -W_{g-2} + (g-1)P + (g-2)Q$$

ending the proof that for the choice (4.5) the intersection $W_{g-1} \cap (W_{g-1} + u - v)$ consists of the union of two identical components; that is to say the intersection multiplicity of D_r and D_{r+1} is two.

Secondly, since

$$D_{i+k}=D_i+k(Q-P)$$

and since for typical $\mathscr{C}$ and $k \neq 1$,

$$k(P-Q) \neq v-u \quad \text{for } u, v \in W_1$$

$D_i \cap D_{i+k}$ is irreducible; in this argument, one must exclude the possibility that

$$\operatorname{Jac}(\mathscr{C}) = \text{Elliptic curve} \oplus \text{another factor, (isogeny)},$$

which would only happen for special $\mathscr{C}$. This proves (i) and (ii), but with D_k in place of D_{α_k}. Hence we have for all $k, j>1$,

$$\operatorname{mult}(D_k \cdot D_{k+1})=2, \qquad \operatorname{mult}(D_k \cdot D_{k+j})=1,$$

while from Theorem 2 it immediately follows that for all k and $j>1$,

$$\operatorname{mult}(D_{\alpha_k} \cdot D_{\alpha_{k+1}})=2, \qquad \operatorname{mult}(D_{\alpha_k} \cdot D_{\alpha_{k+j}})=1, \qquad j>1,$$

and so we must have the identity

$$D_k=D_{\alpha_k},$$

proving

$$D_{\alpha_k}=\Theta+k(Q-P), \qquad k=0,1,\ldots,l.$$

At the same time we have shown (i) and (ii).

We now proceed to prove the two formulas (4.2): By Poincaré's lemma, we have

$$W_d \sim \frac{1}{(g-d)!}\,\Theta^{(g-d)}, \qquad \text{(homologous)};$$

this formula combined with (1.18) (applied to $a_l^{(1)}$) leads to

$$D_{\alpha_{i_1}} \cdot D_{\alpha_{i_2}} \cdot \ldots \cdot D_{\alpha_{i_k}} \sim D_0^k = \Theta^k \sim k!\, W_{l-k}. \tag{4.6}$$

According to Theorem 2,

$$\operatorname{mult.}(D_{\alpha_{i_1}} \cdot \ldots \cdot D_{\alpha_{i_k}}, \alpha_i \in \Delta') = \frac{|W_{\Delta'}|}{\det(A_{\Delta'})} = \prod_r \left(\frac{|W_{\Delta_r''}|}{\det(A_{\Delta_r''})}\right) = \prod_{\Delta_r'' \subset \Delta'} (|\Delta_r''|!) \tag{4.7}$$

with

$$\Delta' = \bigcup_r \Delta_r'' \text{ (disjoint) and } \Delta_r'' \text{ connected (i.e. the root system of some } a_{|\Delta_r''|});$$

then combining (4.6) and (4.7), one finds the following statement for the reduced variety (i.e. not taking into account the multiplicity):

$$\operatorname{reduced}(D_{\alpha_{i_1}} \cdot \ldots \cdot D_{\alpha_{i_k}}, \alpha_i \in \Delta') = \frac{|\Delta'|!}{|\Delta_1''|! \ldots |\Delta_r''|! \ldots}\, W_{l-k}. \tag{4.8}$$

It is interesting to consider the two extremes:

Corollary. 1. *If $\Delta' \subset \Delta$ is indecomposable (i.e. corresponding to some a_k)*

$$\textit{mult.}(D_{\alpha_{i_1}} \cdot \ldots \cdot D_{\alpha_{i_k}}, \alpha_i \in \Delta') = k!$$

$$\textit{reduced}(D_{\alpha_{i_1}} \cdot \ldots \cdot D_{\alpha_{i_k}}) \sim W_{l-k};$$

in particular, if $k=l-1$ *($\hat{D}$ means: D omitted)*

$$\text{(4.8)} \qquad \textit{mult.}\,(D_{\alpha_0}\cdot\ldots\cdot\hat{D}_{\alpha_r}\cdot\hat{D}_{\alpha_{r+1}}\cdot\ldots\cdot D_{\alpha_l})=(l-1)!$$

$$\textit{reduced}\,(D_{\alpha_0}\cdot\ldots\cdot\hat{D}_{\alpha_r}\cdot\hat{D}_{\alpha_{r+1}}\cdot\ldots\cdot D_{\alpha_l})\sim W_1=\mathscr{C}.$$

2. If Δ' consists of totally isolated roots (no two roots of Δ' are connected with a link)

$$\textit{mult.}\,(D_{\alpha_{i_1}}\cdot\ldots\cdot D_{\alpha_{i_k}})=1$$

$$\textit{reduced}\,(D_{\alpha_{i_1}}\cdot\ldots\cdot D_{\alpha_{i_k}})\sim k!\,W_{l-k}.$$

Conjecture: *referring to case 1,*

$$(D_{\alpha_0}\cdot\ldots\cdot\hat{D}_{\alpha_r}\cdot\hat{D}_{\alpha_{r+1}}\cdot\ldots\cdot D_{\alpha_l})=(l-1)!\,W_1=(l-1)!\,\mathscr{C}$$

with intersection multiplicity $(l-1)!$.

For $l=3$, it is precisely statement (i) of Theorem 5. We check it for $l=3$, using the Laurent solutions. At first, notice that

$$\mathscr{C}:\det(\mathrm{L}_h-zI)=h+h^{-1}-P_4(z)=0\,,\qquad H_0=x_1x_2x_3x_4=1$$

with

$$P_4(z)=z^4+6kz^2-2lz+m\,.$$

Secondly, the Laurent solutions $D_{\{\alpha_1,\alpha_2\}}$ have the form:

$$\begin{aligned}
D_{\{\alpha_1,\alpha_2\}}: x_1&=t^{-2}(2+Xt^2+Wt^3+\ldots) & y_1&=t^{-1}(2+Vt-Xt^2-Wt^3/2+\ldots)\\
x_2&=t^{-2}(2+Xt^2-Wt^3+\ldots) & y^2&=V+Wt^2+(U-Y)t^3/9+\ldots\\
x_3&=Ut^2+\ldots & y^3&=t^{-1}(-2+Vt+Xt^2-Wt^3/2+\ldots)\\
x_4&=Yt^2+\ldots & y^4&=-3V+(U-Y)t^3/3+\ldots\,;
\end{aligned}$$

putting them into the constants of motion yields

$$D_{\{\alpha_1,\alpha_2\}}\cap\mathscr{A}=\left\{\begin{matrix} X=V^2+k, & Y=1/4U, & W=-(10V^3+6kU+l)/5\\ & 2U+\dfrac{1}{2U}-P_4(-3V)=0 & \end{matrix}\right\};$$

this curve is precisely $\mathscr{C}$; thus combining with the corollary above, we have $D_{\alpha_1}\cap D_{\alpha_2}=2W_1$.

2. The Toda lattice for $c_n^{(1)}$

The Lie algebra $c_n^{(1)}$ can be viewed as $a_{2n-1}^{(1)}$ with some identifications. Using the standard representation, the Lax pair for the $c_n^{(1)}$-Toda equations is a special case of the previously discussed $a_{2n-1}^{(1)}$-Toda equations, with the following identifications

$$x_i=x_{2n-i}\,,\qquad y_i=-y_{2n+1-i}\,,$$

i.e.,

$$L=\left[\begin{array}{cccccccccc} y_1 & -x_1 & 0 & & & & & & & h^{-1} \\ 1 & y_2 & -x_2 & & & & & & & \\ 0 & 1 & y_3 & & & & & O & & \\ & & & y_{n-1} & -x_{n-1} & 0 & & & & \\ & & & 1 & y_n & -x_n & 0 & & & \\ \hline & & & 0 & 1 & -y_n & -x_{n-1} & & & \\ & & & & 0 & 1 & -y_{n-1} & & & \\ & O & & & & & & & & \\ & & & & & & & -y_3 & -x_2 & 0 \\ & & & & & & & 1 & -y_2 & -x_1 \\ -x_0 h & & & & & & & 0 & 1 & -y_1 \end{array}\right] \tag{4.9}$$

The isospectral curve $\mathscr{C}$ has the following form

$$\mathscr{C}: h+h^{-1}+P_n(z^2)=0\,, \qquad \text{genus}\,\mathscr{C}=g=2n-1\,,$$

and it supports an involution σ, in addition to τ:

$$\tau:(z,h)\to(z,h^{-1})\,, \qquad \sigma:(z,h)\to(-z,h)\,.$$

The curve $\mathscr{C}$ can thus be viewed as a double cover of $\mathscr{C}_0$

$$\mathscr{C}_0: h+h^{-1}+P_n(u)=0\,, \qquad \text{genus}\,(\mathscr{C}_0)=g_0=n-1\,.$$

This leads us to consider the Prym variety; namely

$$\text{Prym}\,(\mathscr{C}/\mathscr{C}_0)=\{\mathscr{D}\in\text{Jac}\,(\mathscr{C}) \quad \text{such that} \quad \mathscr{D}+\mathscr{D}^\sigma\equiv 0 \text{ in } \text{Jac}\,\mathscr{C}\}$$

is an Abelian subvariety of Jac $(\mathscr{C})$ and we have the decomposition

$$\text{Jac}\,(\mathscr{C})=\text{Jac}\,\mathscr{C}_0\oplus\text{Prym}\,(\mathscr{C}/\mathscr{C}_0) \text{ (modulo an isogeny)}$$

with

$$\dim\text{Jac}\,(\mathscr{C}_0)=g_0=n-1 \quad \text{and} \quad \dim\text{Prym}\,(\mathscr{C}/\mathscr{C}_0)=n\,.$$

The affine invariant manifolds $\mathscr{A}_0$ for the $c_n^{(1)}$-Toda equations are given by

$$\begin{aligned}\mathscr{A}_0=\{H_0=x_0x_1^2x_2^2\ldots x_{n-1}^2x_n=1, \mathrm{P}_n(z^2)&\equiv z^{2n}+H_1z^{2n-2}+\ldots+H_n\\ &=z^{2n}+c_1z^{2n-2}+\ldots+c_n\}\,.\end{aligned}$$

The injective map $i:\mathscr{A}\to\text{Jac}\,(\mathscr{C})$ for $a_{2n-1}^{(1)}$, considered in the proof of Theorem 5 and restricted to $\mathscr{A}_0$, leads to an injective map to Prym $(\mathscr{C}/\mathscr{C}_0)$.

Theorem 6. *The Prym variety* Prym $(\mathscr{C}/\mathscr{C}_0)$ *has a polarization given by*

$$(1,2,\ldots,2)\,;$$

moreover the injective map $i:\mathscr{A}\to\text{Jac}\,(\mathscr{C})$ *for* $a_{2n-1}^{(1)}$ *restricted to* $\mathscr{A}_0$ *leads to an injective map*

$$i_0:\mathscr{A}_0\to\text{Prym}\,(\mathscr{C}/\mathscr{C}_0)\subset\text{Jac}\,(\mathscr{C}) \tag{4.10}$$

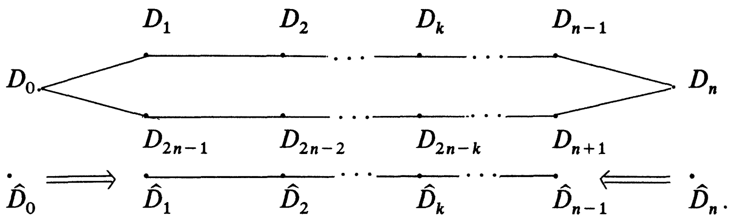
with

$$\operatorname{Prym}(\mathscr{C}/\mathscr{C}_0)\backslash i(\mathscr{A}_0)=\sum_{k=0}^{n}\hat{D}_k,$$

and

$$\hat{D}_k\cap\operatorname{Prym}(\mathscr{C}/\mathscr{C}_0)=D_{2n-k}\cap\operatorname{Prym}(\mathscr{C}/\mathscr{C}_0)\,;$$

the identifications above are compatible with the Dynkin diagram



Moreover, comparing with the Painlevé divisors $D_{\{k\}}(0)$, we have

$$D_{\{k\}}(0)=\hat{D}_k \tag{4.11}$$

and the divisors $\hat{D}_0$ and $\hat{D}_n$ are isomorphic.

Remark. Indeed, the circular Dynkin diagram of $a_{2n-1}^{(1)}$ can be collapsed to the one of $c_n^{(1)}$. This matches the identification of the x_i's in going from $c_n^{(1)}$ to $a_{2n-1}^{(1)}$ in the matrix $\hat{L}_h$ in (4.1). The claim is that the corresponding divisors D_k and D_{2n-k}, in the $a_{2n-1}^{(1)}$ picture, intersect the Prym variety according to one and the same divisor $\hat{D}_k$ in Prym.

Proof. To show the validity of (4.10), we inspect the divisor of poles of the eigenvector $f=(f_1,\dots,f_{2n})$ of $L_hf=zf$, normalized at $f_{2n}=1$. Indeed

$$f_i=\frac{\Delta_{Ni}}{\Delta_{NN}}=\frac{\Delta_{ii}}{\Delta_{iN}}\quad i=1,\dots,N=2n\,.$$

Since, by the symmetries of the matrix (4.9), $\Delta_{N1}^{\tau}=-\Delta_{1N}$, $\Delta_{ii}^{\tau}=\Delta_{ii}$, $\Delta_{NN}^{\sigma}=-\Delta_{11}$, their divisors are as follows:

$$\begin{aligned}&(\Delta_{1N})=\mathscr{D}+\mathscr{D}^{\sigma}-2gP && (\Delta_{N1})=\mathscr{D}^{\tau}+\mathscr{D}^{\tau\sigma}-2gQ\\ &(\Delta_{NN})=\mathscr{D}+\mathscr{D}^{\tau}-gP-gQ && (\Delta_{11})=\mathscr{D}^{\sigma}+\mathscr{D}^{\tau\sigma}-gP-gQ\end{aligned} \tag{4.12}$$

which implies

$$(f_1)=\mathscr{D}^{\tau\sigma}-\mathscr{D}-gQ+gP,$$

and using the divisor (Δ_{1N}), conclude

$$\int_{gP}^{\mathscr{D}}\omega+\int_{gP}^{\mathscr{D}^{\sigma}}\omega=0,$$

establishing the range of the map $\mathscr{A}_0\to\operatorname{Prym}(\mathscr{C}/\mathscr{C}_0)$.

Since the curves $\mathscr{C}_0$ and $\mathscr{C}$ have genera $g_0=n-1$ and $g=2n-1$ respectively, we have that (for an elementary proof see L. Haine [14]) the polarization of $\operatorname{Prym}(\mathscr{C}/\mathscr{C}_0)$ is given by

$$(\overbrace{1,,\ldots,1}^{g-2g_0},\overbrace{2,,\ldots,2}^{g_0})=(1,\overbrace{2,\ldots,2}^{n-1})$$

in agreement with the results announced in Theorem 6.

In view of the injective map (4.10), we have that

$$\begin{aligned}\operatorname{Prym}(\mathscr{C}/\mathscr{C}_0)\backslash i(\mathscr{A}_0)&=(\operatorname{Jac}(\mathscr{C})\backslash i(\mathscr{A}))\cap\operatorname{Prym}(\mathscr{C}/\mathscr{C}_0)\\&=\left(\sum_0^{2n-1} D_k\right)\cap\operatorname{Prym}(\mathscr{C}/\mathscr{C}_0)\\&=\sum_0^{2n-1}(D_k\cap\operatorname{Prym}(\mathscr{C}/\mathscr{C}_0))\equiv\sum_0^{2n-1}\hat{D}_k,\end{aligned}$$

with the following identifications,

$$\{x_k^{-1}=0\}\cap\operatorname{Prym}=\hat{D}_k\leftrightarrow\hat{D}_{2n-k}=\{x_{2n-k}^{-1}=0\}\cap\operatorname{Prym}$$

compatible with the Dynkin diagram and generated by the identifications $x_k=x_{2n-k}$.

The proof of the remaining statements, concerning the Painlevé divisors, involve some computations which we perform in the case $n=2$. Thus from the matrix L above, we find the isospectral curve

$$\mathscr{C}=\left\{\begin{aligned}\det(zI-L_h)&=-(h+h^{-1})+z^4+(-y_1^2-y_2^2+x_0+2x_1+x_2)z^2\\&\qquad+((y_1^2-x_0)(y_2^2-x_2)+2x_1(x_1+y_1y_2))=0\\&\equiv-(h+h^{-1})+z^4+kz^2+l=0\end{aligned}\right\}$$

and from (4.12) it follows that the following divisor on $\mathscr{C}$ consists of 3 points:

$$\mathscr{D}=\{\Delta_{44}=\Delta_{14}=0\text{ on }\mathscr{C}\}=\sum_1^3\{z_i,h_i\}$$

where z_i are the three roots of $\Delta_{44}(z)=0$ and

$$h_i=\frac{1}{x_0(z_i^2+x_2-y_2^2)}=-\frac{(z_i-y_1)}{x_0x_1(z_i+y_2)}\qquad i=1,2,3.$$

The latter follows from setting $\Delta_{14}=0$ and then using $\Delta_{44}=0$, where

$$\begin{aligned}&\Delta_{44}(z,h)=-z^3+y_1z^2-(x_1+x_2-y_2^2)z-(-y_1x_2+y_2x_1+y_1y_2^2)\\&\Delta_{11}(z,h)=-\Delta_{44}(-z,h)\\&\Delta_{14}(z,h)=x_0h(-z^2+y_2^2-x_2)+1\\&\Delta_{41}(z,h)=-\Delta_{14}(z,h^{-1}).\end{aligned}$$

The Laurent solutions $D_{\{0\}}(t)$, $D_{\{1\}}(t)$ and $D_{\{2\}}(t)$ read as follows:

$D_{\{0\}}(t)$	$D_{\{1\}}(t)$	$D_{\{2\}}(t)$
$x_0=\frac{1}{t^2}+X+\dots$	$x_0=Yt^2(1+2Vt)+\dots$	$x_0=Y+2VYt+\dots$
$x_1=-Ut(1+Vt+\dots)$	$x_1=\frac{1}{t^2}+X+\dots$	$x_1=Ut(1-Vt+\dots)$
$x_2=Y-2VYt+\dots$	$x_2=Ut^2(1-2Vt)+\dots$	$x_2=\frac{1}{t^2}+X+\dots$
$y_1=-\frac{1}{t}+tX+\dots$	$y_1=\frac{1}{t}+V-tX+\dots$	$y_1=V+tY+\dots$
$y_2=V+tY+\dots$	$y_2=-\frac{1}{t}+V+tX+\dots$	$y_2=\frac{1}{t}-tX+\dots$
	$D_{\{0\}}(0):\left\{\begin{matrix}U^2Y=1,\ Y+3X-V^2=k\\ 3XY-3V^2X+2UV=l\end{matrix}\right\}$	
	$D_{\{1\}}(0):\left\{\begin{matrix}UY=1,\ 3X=V^2+k\\ U+U^{-1}=4V^4+8kV^2+4k^2-l\end{matrix}\right\}$	
	$D_{\{2\}}(0):\left\{\begin{matrix}\text{same formula}\\ \text{as } D_{\{0\}}(0)\end{matrix}\right\}$	

We now show that

$$D_{\{0\}}(0)=\hat{D}_0 \quad D_{\{1\}}(0)=\hat{D}_1 \quad D_{\{2\}}(0)=\hat{D}_2 .$$

Indeed, the points $(x_0(t), x_1(t), x_2(t), y_1(t), y_2(t))$, $t\neq 0$ given by the Laurent series above belong to the affine variety $\mathscr{A}_0$. We now compute the divisor in Jac$(\mathscr{C})$, by substituting the series for x and y into Δ_{44} and Δ_{14} and by solving for z. One finds

$$-\Delta_{44}(z)|_{D_{\{\alpha_0\}}(t)}=\prod_1^3 (z-z_i(t))=\left(z+\frac{1}{t}\right)(z^2+Y-V^2)+\dots=0 .$$

Therefore we have

$$z_1(t)=-\frac{1}{t}+\dots,\qquad z_2(t)=\sqrt{V^2-Y}+\dots,\qquad z_3(t)=-\sqrt{V^2-Y}$$

$$h(z_i(t))|_{D_{\{\alpha_0\}}(t)}=\left.\frac{1}{x_0(-z_i^2(t)+y_2^2-x_2)}\right|_{D_{\{\alpha_0\}}(t)}=-\frac{1}{x_0x_1}\left.\left(\frac{z_i(t)-y_1}{z_i(t)+y_2}\right)\right|_{D_{\{\alpha_0\}}(t)} .$$

Using the first expression for $z_1(t)$ and the second for z_2 and z_3, yields

$$h(z_i(t))|_{D_{\{\alpha_0\}}(t)}\begin{cases}=t^4+\dots & \text{for } i=1\\[4pt] =\dfrac{1}{U(-V-\sqrt{V^2-Y})}+\dots & \text{for } i=2\\[4pt] =\dfrac{1}{U(-V+\sqrt{V^2-Y})}+\dots & \text{for } i=3\,;\end{cases}$$

letting $t\to 0$, the divisor on $\mathscr{C}$ has the following form

$$Q+\left(\sqrt{V^2-Y},\frac{1}{U(-V-\sqrt{V^2-Y})}\right)$$
$$+\left(-\sqrt{V^2-Y},\frac{1}{U(-V+\sqrt{V^2-Y})}\right)\in Q+S^2\mathscr{C}.$$

Since $\mathscr{A}_0$ maps to Prym $(\mathscr{C}/\mathscr{C}_0)$, this point belong to Prym $(\mathscr{C}/\mathscr{C}_0)$ as well. Therefore, remembering $D_0=Q+W_2$, we have shown $D_{\{0\}}(0)=\hat{D}_0$.

Next we show that $D_{\{1\}}(0)=\hat{D}_1$. Indeed, a point in $\mathscr{A}_0$ near $D_{\{1\}}(0)$ is mapped into a divisor in Jac $(\mathscr{C})$, specified by the roots of

$$\Delta_{44}(z)\Big|_{\substack{x(t),\,y(t)\text{ as}\\ \text{in }D_{\{1\}}(t)}}=z^3-z^2\left(\frac{1}{t}+V\right)+\left(\frac{2V}{t}+X-V^2\right)z+\left(-\frac{1}{t}+V\right)(V^2+X)$$
$$+\text{lower order term } s=0$$
$$=\left(z-\frac{1}{t}+V\right)(z^2-2Vz+(V^2+3X))+\text{lower order term } s=0$$

leading to

$$z_1=\frac{1}{t}-V+\dots,\qquad z_2=V+\sqrt{-3X}+\dots,\qquad z_3=V-\sqrt{-3X}+\dots$$

and

$$h(z_1)=\frac{c}{t^2}+\dots,\qquad h(z_2)=U+\dots,\qquad h(z_3)=U+\dots.$$

Taking the limit when $t\to 0$, the divisor $\sum_1^3 (z_i,h(z_i))$ tends to

$$P+(V+\sqrt{-3X},U)+(V-\sqrt{-3X},U)\in P+S^2\mathscr{C}.$$

Since again $\mathscr{A}_0$ maps to Prym $(\mathscr{C}/\mathscr{C}_0)$, this point belongs to Prym $(\mathscr{C}/\mathscr{C}_0)$ as well. Therefore, remembering $D_{-1}=P+W_2$, we have

$$D_{\{1\}}(0)=\hat{D}_{-1}\equiv\hat{D}_3=D_1.$$

To obtain (iii), one uses the map in $a_3^{(1)}$ given by shifting the indices by 2; this is an involution in $a_3^{(1)}$:

$$(x_0,x_1,x_2,x_3,y_1,y_2,y_3,y_4)\to(x_2,x_3,x_0,x_1,y_3,y_4,y_1,y_2),$$

which induces the interchange of divisors in $a_3^{(1)}$:

$$D_0\leftrightarrow D_2,\qquad D_1\leftrightarrow D_3.$$

Since $c_2^{(1)}$ is obtained by factoring $a_3^{(1)}$, as explained earlier, the involution above factors into

$$(x_0,x_1,x_2,y_1,y_2)\leftrightarrow(x_2,x_1,x_0,-y_2,-y_1)$$

and at the levels of the divisors $\hat{D}_i$:

$$\hat{D}_0\leftrightarrow\hat{D}_2,\qquad \hat{D}_1\leftrightarrow\hat{D}_1.$$

Therefore $D_{\{2\}}(0) = \hat{D}_2$; but since the curves $D_{\{2\}}(0)$ and $D_{\{0\}}(0)$ are isomorphic, the curves $\hat{D}_0$ and $\hat{D}_2$ are isomorphic as well, concluding the proof of Theorem 6.

Remark. By substituting the Laurent solutions $D_{\{\alpha_1, \alpha_3\}}(t)$ of $\mathscr{a}_3^{(1)}$ into the 4 constants of motion and then specializing to $\mathscr{c}_2^{(1)}$ by setting the invariant of degree 3 equal to zero, one finds the following intersection

$$D_1 \cap D_3 = D_{\{1\}}(0) \cup (\text{elliptic curve}),$$

to be reducible, whereas the fact that

$$D_3 = D_1 + 2(Q-P) \neq D_1 + u - v \quad \text{with} \quad u, v \in W_1$$

seems to indicate that $D_1 \cap D_3$ would be irreducible (see Gunning [13]). Gunning's theorem states that for a Riemann surface of genus $g \geqq 5$ ($g=3$ in our case), the variety $W_{g-1} \cap (W_{g-1} + X)$ has two components only if $X \in W_1 - W_1$, or one of the components has the form $W+T$; X is an irreducible subvariety of dimension $g-3$ (a point in our case) and T is an elliptic subvariety.

References

1. Adler, M., Moerbeke van, P.: Kowalewski's Asymptotic Method, Kac-Moody Lie Algebras, and Regularization. Commun. Math. Phys. **83**, 83–106 (1982)
2. Adler, M., Moerbeke van, P.: The complex geometry of the Kowalewski-Painlevé analysis. Invent. Math. ⁸7, 3–51 (1989)
3. Adler, M., Moerbeke van, P.: Algebraic Complete integrable systems: A systematic approach (1985). Perspect. Math. 1989 (to appear)
4. Arnolds, V.I.: Singularity theory (selected papers). London Math. Soc. Lect. Notes **53**
5. Arnold, V.I., Gusein-Zade, S.M., Varchenko, A.N.: Singularities of differentiable maps, vol. I, monographs in math. Boston: Birkhäuser 1985
6. Bogoyavlenski, O.I.: On perturbartions of the periodic Toda lattices, Commun. Math. Phys. **51** (1976)
7. Bourbaki, N.: Groupes et algêbres de Lie, chap. 4, 5, 6 (Eléments de mathématique. Paris: Hermann 1968
8. Brieskorn, E.: Singular elements of semi-simple algebraic groups. Actes Congr. Int. Math. **2**, 279–284 (1970)
9. Adler, M., Haine, L., Moerbeke van, P.: Limit operators for the Toda flow and periodic flags for Loop groups. (to appear)
10. Flaschka, H.: The Toda lattice in the complex domain. Prospect of Algebraic Analysis (to appear)
11. Flaschka, H., Zeng, Y.: Painlevé Analysis for the semi-simple Toda lattice (preprint)
12. Griffiths, Ph., Harris, J.: Principles of Algebraic Geometry. New York: Wiley-Interscience 1978
13. Gunning, R.C.: Lectures on Riemann surfaces, Jacobi varieties, Princeton: Princeton University Press 1972
14. Haine, L.: Geodesic flow on $SO(4)$ and Abelian surfaces. Math. Ann. **263**, 435–472 (1983)
15. Helgason, S.: Differential geometry, Lie groups and symmetric spaces. New York, London: Academic Press 1978
16. Humphreys, J.G.: Introduction to Lie algebras and Representation theory. New York Berlin Heidelberg: Springer 1972
17. Kac, V.G.: Infinite Dimensional Lie algebras. Boston: Birkhäuser 1984
18. McKean, H.P.: Integrable system and algebraic curves, Global analysis (Lect. Notes Math., vol. 755, pp. 83–200) Berlin Heidelberg New York: Springer 1979
19. Mumford, D.: Algebraic geometry I, Complex projective varieties, Berlin Heidelberg New York: Springer 1976
20. Mumford, D.: Curves and their Jacobians. The University of Michigan Press 1975

21. Mumford, D.: Tata lectures on theta II (Progr. Mathematics, vol. 43) Boston: Birkhäuser 1984
22. Ol'shanetsky, A., Perelomov, A.M.: Explicit solutions of classical generalized Toda models. Invent. Math. **54**, 261–269 (1979)
23. Moerbeke van, P., Mumford, D.: The spectrum of difference operators and algebraic curves. Acta Math. (1979)
24. Yoshida, H.: Integrability of generalized Toda lattice systems and singularities in the complex plane in Non-integrable Systems – Classical Theory and Quantuum Theory, Proc. RIMS Colloquium World Scientific, Singapore 1983
25. Adler, M., Moerbeke van, P.: Limit of differential operators and theta functions, from the Grassmannian point of view (to appear)
26. Goodman, R., Wallach, N.: Classical and quantum-mechanical systems of Toda lattice type I. Comm. Math. Phys. **83**, 355–386 (1982)
27. Kostant, B.: On Whittaker vectors and representation theory. Invent. Math. **48**, 101–184 (1978)
28. Pressley, N., Segal, G.: Loop groups. Clarendon Press, Oxford 1986
29. Reyman, A.G., Semenov-Tian-Shansky, M.A.: Reduction of Hamiltonian Systems. Affine Lie algebras and La equations. I, II, Invent. Math. **54**, 81–100 (1979)